Molecular Therapy
Oncology
Original Article

Targeted expansion of cytotoxic T cells using IL-12 and CD137L supplementation enhances antitumor efficacy

Marta Sanz,[1] Brendan T. Mann,[1] Elyse K. McMahon,[1] Alberto Bosque,[1] Samuel Simmens,[2] and Natalia Soriano-Sarabia[1]

[1]Departments of Microbiology Immunology and Tropical Medicine, George Washington University, Washington, DC 20037, USA; [2]Biostatistics, George Washington University, Washington, DC 20037, USA

The increasing success of allogenic Vδ2 T cell immunotherapy for the treatment of cancer has been demonstrated in recent studies. Vδ2 T cells recognize phosphoantigens, intermediates of the mevalonate pathway, through butyrophilin molecules, and they are not major histocompatibility complex (MHC) restricted. Allogeneic transfer of *in vitro* expanded Vδ2 T cells has shown more promising results than autologous strategies, although the clinical benefit remains limited. One of the issues leading to less-than-optimal responses relates to the polyclonal expansion of Vδ2 T cell subsets with variable cytotoxic capacity. Previous work developed protocols to expand Vδ2 T cells, although to our knowledge, ours is the first comprehensive study that has produced a simple, antigen-presenting feeder-free culture that produced an average expansion of 3,000-fold and more than 95% pure Vδ2 T cells avoiding additional isolation steps. Here, we show the *in vitro* expansion of cytotoxic Vδ2 T cells expressing CD16 and NKG2A enriched in granzyme B that displayed enhanced antitumor activity of up to 40% against leukemia and ovarian, breast, and lung cancer cells. Our work warrants clinical testing to evaluate the therapeutic potential of these highly cytotoxic cells, paving the way for improved efficacy of personalized cell-based immunotherapies.

INTRODUCTION

The most abundant peripheral subset of γδ T cells, Vδ2 T cells, recognizes nonpeptidic phosphorylated intermediate metabolites or phosphoantigens (P-Ags) of the mevalonate (MVA) pathway such as isopentenyl pyrophosphate (IPP).[1–4] Intracellular concentrations of IPP are increased due to malignancy or infection and induce specific Vδ2-T cell receptor (TCR) activation mediated by butyrophilin molecules.[5–7] In addition, aminobisphosphonates (N-BPs) such as alendronate (ALN), pamidronate (PAM), or zoledronate (ZOL) are compounds that modulate the MVA pathway, leading to increased IPP production and activation of Vδ2 T cells.[1,8,9] IPP interacts with the B30.2 (or CD277) intracellular domain of butyrophilin molecule BTN3A1 that associates with BTN2A1 inducing TCR-γδ signaling.[6,7,10] Activated Vδ2 T cells are potent cytotoxic effectors against different types of malignancies[11–14] and exert their cytotoxic functions through the release of perforin, granzymes (Gzm), and interferon-γ (IFN-γ) or by inducing death receptor-mediated apoptosis such as tumor necrosis factor (TNF)-related apoptosis-inducing ligand or Fas ligand (FasL), members of the TNF receptor superfamily[12,15–17] Cytotoxic Vδ2 T cells can be identified by the expression of markers such as NKG2D, CD56, NKG2A, or CD16, as we and others previously described.[18–22]

NKG2A is an inhibitory receptor with key roles in the regulation of γδ T cells, αβ-CD8$^+$ T cells, and the effector functions of natural killer (NK) cells[18,23–25] that bind to the non-classical class I human leukocyte antigen E (HLA-E).[26,27] Overexpression of HLA-E is a tumor-escape mechanism that limits the effector functions of tumor-infiltrating lymphocytes, and therefore, the major histocompatibility complex (MHC)-like NKG2A/HLA-E axis plays a critical role in antitumor immunity.[28,29] Targeting the NKG2A/HLA-E axis has been shown to enhance the antitumor activity of cytolytic effector populations.[18,25,30] In addition to the well-recognized involvement of CD16 exerting antibody-dependent cellular cytotoxicity,[31–33] CD16 also identifies Vδ2 T cells that display a highly cytotoxic phenotype against both cancer cell lines and HIV-infected targets[22,34–37] and has recently provided pre-clinical evidence of its importance as a biomarker.[22]

The MHC-unrestricted nature of Vδ2 T cell antigen recognition[5–7,11,17] represents an advantage over other cell types to develop allogeneic immunotherapies as shown by the increasing number of clinical studies demonstrating safety, efficacy, and tumor-infiltrating capabilities.[22,38–42] However, clinical success remains limited despite the recent success against several advanced malignancies.[41] Reasons for this limited success include donor variability, potential short-lived

Received 20 February 2025; accepted 12 May 2025;
https://doi.org/10.1016/j.omton.2025.200996.

Correspondence: Natalia Soriano-Sarabia, Departments of Microbiology Immunology and Tropical Medicine, George Washington University, Washington, DC 20037, USA.
E-mail: nataliasorsar@gwu.edu

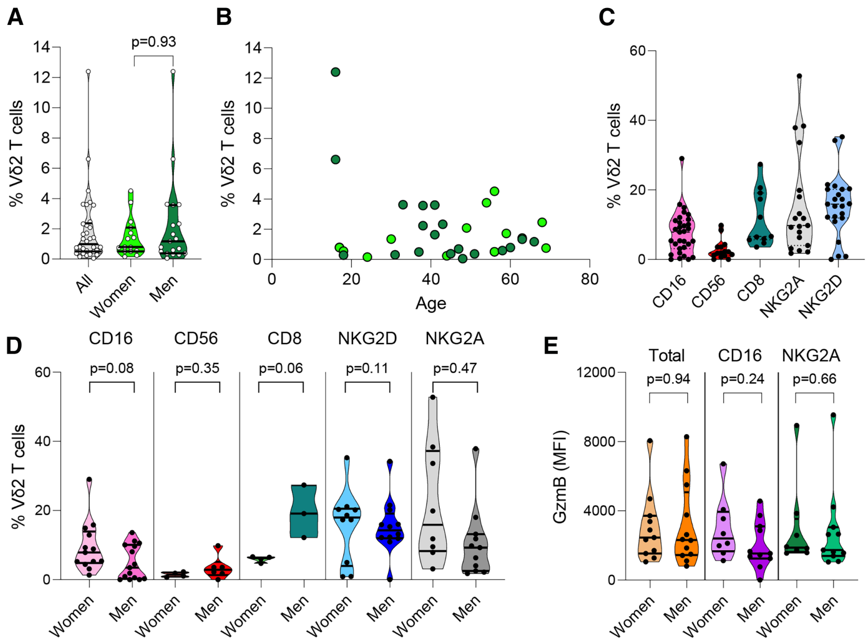

Figure 1. Phenotypic characterization of Vδ2 T cells

(A) Frequency of Vδ2 T cells in the 38 donors included in the study and comparison by self-reported sex. (B) Lack of correlation between the frequency of Vδ2 T cells and age (Spearman test). Distribution of surface markers' expression on Vδ2 T cells (C) in the entire cohort and (D) compared by sex (Mann-Whitney U test). CD16 (n = 31), CD56 (n = 15), CD8 (n = 11), NKG2A (n = 19), and NKG2D (n = 23). (E) Distribution of intracellular GzmB production in women and men. Violin plots represent the distribution where the median and quartiles are represented by black lines.

persistence, incomplete tumor infiltration, and an immunosuppressive tumor environment.[22,42,43] In particular, most of the clinical studies have utilized ZOL and interleukin (IL)-2 as the methods to expand Vδ2 T cells *in vitro*, missing the opportunity to alter their phenotype and function during culture.[42,44] An approach using cytokines combined with other compounds could help develop an optimized protocol to generate Vδ2 T cells with enhanced capacity to target and kill malignant cells from different origin.[42] In this regard, previous methods have included combinations of N-BPs (PAM or ZOL) or synthetic P-Ags with additional cytokines like IL-15, IL-12, IL-21, or IL-18[21,45–54] and compounds like vitamin C.[55,56] These improved protocols have achieved greater Vδ2 T cell numbers with increased expression of CD16, CD56, Nkp44, Nkp30, NKG2A, IFN-γ, perforins, or GzmB.[21,47,49,57–59] Additional manipulation during the expansion period using a genetically modified K562 cell line CD3scFv/CD137L/CD28scFv/IL15RA quadruplet artificial antigen-presenting cells (APCs) enhanced the expansion up to 633-fold.[60] However, simplified and less expensive methods are desirable.

To our knowledge, a comprehensive analysis to test the effect of different N-BPs in combination with other compounds and/or cytokines to selectively target the expansion of cytotoxic Vδ2 T cells with increased antitumor function is lacking. In this study, we report the targeted expansion of highly cytotoxic Vδ2 T cells with improved and broad antitumor activity, using an optimized method.

RESULTS

Characteristics of the study donor volunteers

Buffy coats from 38 healthy volunteer donors were used for the entire study (Table S1). Donors had a median age of 43 (range 16–69) years. Self-reported sex was available for 34 donors, and 15 donors (44%) were women while 19 (56%) were men. Basal frequencies and cytotoxic phenotype of Vδ2 T cells were analyzed by flow cytometry. Among total $CD3^+$ T cells, a median of 1.95% (range 0.05–12.40) were Vδ2 T cells and had a similar representation between women and men (Figure 1A). We did not detect a decline of Vδ2 T cell frequency with age in this cohort (Figure 1B), most likely due to the absence of an even distribution of ages among the donors, which was observed in our previous study.[61] Among Vδ2 T cells, the median basal frequency of CD16 expression was 7.9%, 1.9% for CD56, 6.6% for CD8, 15.9% for NKG2D, and 9.8% for NKG2A (Figure 1C), and these markers were similarly distributed in women and men, except for a tendency of higher levels of CD8 expression among men (p = 0.06, Figure 1D). The distribution of $GzmB^+$ Vδ2 T cells was also comparable between women and men (Figure 1E). Finally, aging was not associated with the frequency of any of the populations expressing CD16, CD8, CD56, NKG2D, NKG2A, or GzmB production (not shown).

ALN promotes enhanced Vδ2 T cell expansion

Previous studies used either ZOL or PAM, but ALN was not previously assayed, and a systematic comparison of their capacity to expand Vδ2 T cells was lacking.[42] To compare the potency of different N-BPs at expanding Vδ2 T cells, 1×10^6 peripheral blood mononuclear cells (PBMCs) from eight healthy donor volunteers were incubated in the presence of eight different concentrations of the N-BPs PAM, ZOL, or ALN (Table 1) in combination with fixed 500 U/mL concentration of IL-2 in complete RPMI (cRPMI) for 10 days. Vδ2 T cell frequency and expression of cytotoxic markers CD16, CD56, and CD8 were analyzed by flow cytometry (Figure S1). Our results showed that a concentration of 3 μM ALN induced the highest Vδ2 T cell expansion (median of 70.3%) compared to all the other tested ALN concentrations, and ZOL (highest median of 54.9% at 2.5 μM) or PAM (highest median of 33.0% at 9 μM) (Figure 2A; Table S2). Interestingly, the concentrations tested in our study reached a plateau at a certain concentration (Figure 2A) despite comparable viability (Figure 2B). The expression of cytotoxic markers CD16, CD56, and CD8 (Figure S2) was variable among the different donors, and the use of diverse N-BPs

Table 1. Tested concentration of N-BPs

Concentration #	PAM (μM)	ZOL (μM)	ALN (μM)
1	1.8	0.5	0.075
2	3.6	1	0.15
3	5.3	1.5	0.3
4	7	2	0.6
5	9	2.5	1.5
6	18	5	2
7	36	10	3
8	90	25	4.5

did not have a measurable impact. These results show that ALN, not commonly used for Vδ2 T cell expansion *in vitro*, induces the highest expansion and should be considered for clinical expansion protocols. In addition, N-BPs and IL-2 induce polyclonal expansion of Vδ2 T cell subpopulations without modifying their cytotoxic profile, missing a unique opportunity to select for highly cytotoxic populations.

Depletion of αβ T cells at the beginning of the culture increases Vδ2 T cell frequency

We previously reported that αβ-TCR$^+$ cells were not required to expand Vδ2 T cells, and they achieved the highest expansion when cocultured with APCs.[62] In addition, other protocols have been more successful when αβ-TCR$^+$ cells were depleted half-way through the expansion protocol.[60,63] Therefore, we tested whether starting the expansion from αβ-TCR$^+$-depleted PBMC would improve Vδ2 T cell frequency upon exposure to 3 μM ALN and 500 U/mL IL-2, which induced the highest median Vδ2 T cell frequency (Figure 2A). αβ-TCR$^+$ T cells were magnetically depleted from PBMCs using samples from nine healthy donor volunteers and were assayed in parallel to whole PBMCs for 10 days. Our results showed that depletion of αβ-TCR$^+$ cells prior to exposure to ALN and IL-2 induced greater Vδ2 T cell expansion compared to total PBMC, with a mean Vδ2 T cell frequency of 86.1% and 53.8%, respectively (Figure 2C). We further confirmed the effect of αβ-TCR$^+$ cells upon exposure to 3 μM ALN that increased the frequencies from a median of 41.8% to 70.0% (Figure S3). Depletion of αβ-TCR$^+$ T cells did not modify the cytotoxic profile of expanded Vδ2 T cells in terms of CD16 expression or GzmB and IFN-γ production (Figure S3).

Finally, we analyzed the distribution of effector/memory Vδ2 T cell populations from αβ-TCR-depleted PBMCs during a culture period of 30 days. We observed that, by day 10, most of the cells had acquired an effector memory (CD27$^-$CD28$^+$) phenotype that continued to increase until day 30 (Figure 2D) without impacting the viability (Figure 2E). The frequency of central memory (CD27$^+$CD28$^-$) cells increased by day 10 and then stabilized, while the proportion of naive cells (CD27$^+$ CD28$^+$) gradually decreased. Terminal differentiated (CD27$^-$CD28$^-$) cells' frequency peaked at day 5 before shrinking and was maintained through day 30 (Figure 2D).

IL-12 and CD137L supplementation enhance the cytotoxic profile of expanding Vδ2 T cells

Previous studies have shown benefits when additional cytokines like IL-12, IL-18, and IL-21 were added to the expansions.[50,55,58,59,64–68] We evaluated which cytokine combinations would provide the greatest frequency of cytotoxic Vδ2 T cell populations. The combination of IL-2 with IL-12 and IL-21 produced the highest CD16 expression compared to IL-18 (Figure S4A) and was used in further combination experiments. In addition, multiple studies have shown that vitamin C, CD277, UMCD6, and CD137L added to the culture may increase the cytotoxic function of Vδ2, NK, and CD8 T cells.[6,7,55,69–72] During the optimization of the culture conditions, our results showed that continuous incubation with either UMCD6 or CD137L in the presence of IL-12 inhibited Vδ2 T cell expansion without compromising the viability (Figures S4B and S4C). Therefore, cultures were supplemented with cytokines during the last 48 h of expansion (Figure 3A). In samples from seven donors, PBMCs were depleted of αβ-TCR$^+$ T cells and incubated in the presence of 3 μM ALN and 500 U/mL IL-2 in cRPMI for 10 days, and IL-2 was refreshed every 2–3 days. At day 5 of culture, all the conditions were exposed to a second dose of 3 μM ALN. ALN and IL-2 culture conditions were compared to ALN+IL-2 combined with either 50 ng/mL vitamin C or 0.5 μg/mL CD277 added on the first day of culture.

The combination of ALN+IL-2 with vitamin C induced the highest Vδ2 T cell frequency, viability, and frequency of Vδ2 CD16$^+$ cells (Figures S5A–S5C). In addition, when combined with IL-12, vitamin C induced the highest expression of NKG2A and CD56 (Figure S5C), while ALN+IL-2 alone induced the highest expression of NKG2D (Figure S5). Our results confirm the beneficial effects of vitamin C promoting Vδ2 T cell expansion, as reported previously.[55,73] However, ALN combined with IL-2+IL-12+CD137L induced the highest frequency of cytotoxic effector Vδ2 T cells compared to all the other conditions (Table S3) analyzed by the intracellular production of GzmB (Figure 3B) and IFN-γ (Figure 3C). GzmB production was primarily produced by CD16$^+$ Vδ2 T cells (Figure 3D) and NKG2A$^+$ Vδ2 T cells (Figure 3E). These results show that the combination of ALN, IL-2, IL-12, and CD137L in αβ-TCR-depleted PBMCs acts synergistically to expand cytotoxic Vδ2 T cell populations.

Selective expansion of cytotoxic Vδ2 T cell phenotypes with IL-12 and CD137L supplementation

To assess whether the optimized protocol using ALN+IL-2+IL-12+CD137L selectively increased the absolute number of cytotoxic Vδ2 T cells, cultures were performed during extended periods, transferring the cells to bigger containers, as required (Figure 4A). Samples from 19 healthy donor volunteers were split in subsequent cytotoxicity experiments (Table S1). First, we analyzed the phenotype and cytolytic function of expanded Vδ2T cells generated with our protocol compared to ALN+IL-2. Our cultures originally contained an average of 2.3% Vδ2 T cells within αβ-TCR-depleted PBMCs. The absolute number of Vδ2 T cells increased from a mean of 1.8×10^5

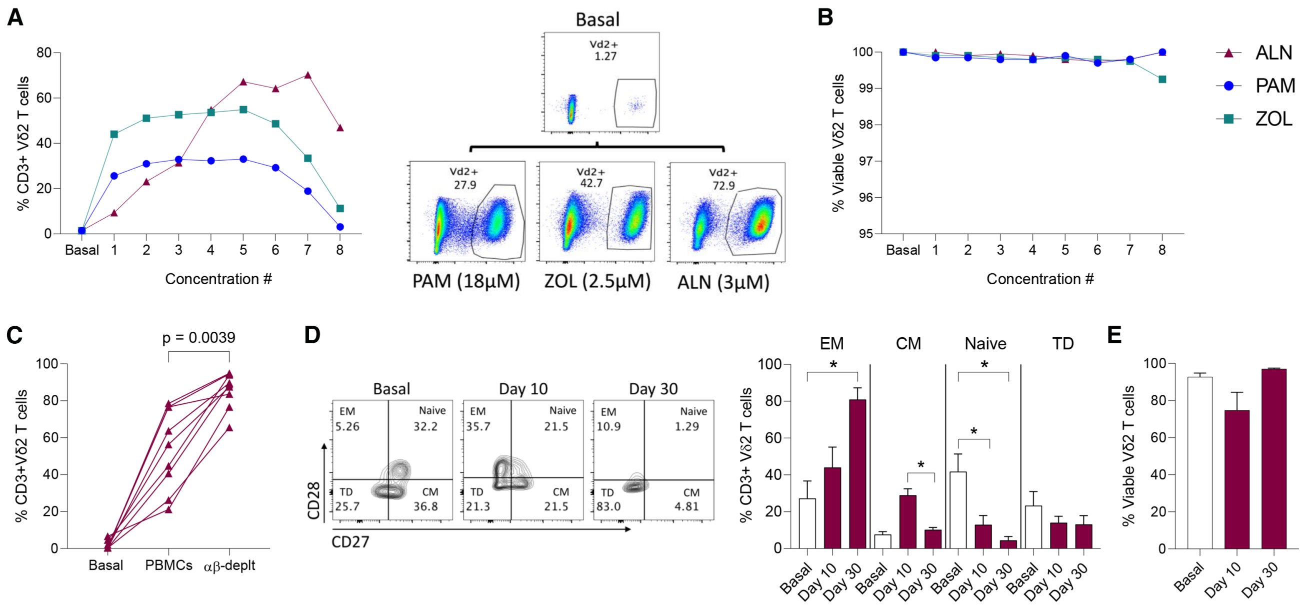

Figure 2. Alendronate and the depletion of αβTCR+ T cells increase Vδ2 T cell expansion in cultures of 10 days

(A) Frequency of Vδ2 T cells upon exposure to increasing concentrations of N-BPs (shown in Table 1) and representative plots. Medians of eight donors are shown. (B) Viability of the Vd2 T cells during the 10-day culture period. (C) Comparison of Vd2 T cell frequencies expanded with 3μM ALN and 500U/mL IL-2 either from total PBMCs or αβ-TCR depleted PBMCs. (D) Modulation of naïve and memory Vd2 T cell populations and (E) viability of expanded populations after 10 and 30 days of expansion. Wilcoxon matched-pairs signed rank tests. Asterisks represent p values < 0.05. CM, central memory. EM, effector memory. TD, terminally differentiated.

cells to 57.1×10^6 cells (Figure 4B), with an average purity of 94.1%. This resulted in a mean fold expansion of 3,105, reaching up to 17,000-fold expansion (Figure 4C) over an average culture period of 19 days (Table 2). The expanded frequency of Vδ2 T cells positively correlated with the basal Vδ2 T cell frequency (Figure 4D).

GzmB production was higher in Vδ2 T cells generated with our optimized protocol compared to ALN+IL-2 (mean fluorescence intensity 28,050 vs. 16,706, respectively, Figure 4E). The frequency of CD16 expression was comparable between the two protocols (Figure 4F), while NKG2A expression was increased compared to ALN+IL-2 (Figure 4G). However, Vδ2 T cells generated with ALN+IL-2+IL-12+CD137L and expressing either CD16 or NKG2A were enriched in GzmB compared to those expanded using ALN+IL-2 (Figures 4H and 4I), confirming our results from short-term cultures of 10 days (Figures 3D and 3E).

Enhanced and broad antitumor function of Vδ2 T cells expanded with an optimized protocol

We assessed whether our optimized protocol using ALN+IL-2+IL-12+CD137L (Figure 4A), compared to ALN+IL-2, improved Vδ2 T cell cytotoxic function against several cancer cell lines—leukemia K562, ovarian A2780, lung H1299 and breast cell lines MCF7 and T47D—using both a non-radioactive cytotoxicity assay and a flow-based method. Target and effector-expanded Vδ2 T cells were cocultured at a 1:1 ratio for 4 or 18 h, respectively, to assess both early and late cytolytic function.

Early cytotoxicity

Our results showed increased killing of Vδ2 T cells expanded with ALN+IL-2+IL-12+CD137L compared to ALN+IL-2 against all of the cell lines analyzed including K562 (mean 64% vs. 37%, Figure 5A), A2780 (mean 39% vs. 20%, Figure 5B), H1299 (mean 65% vs. 24%, Figure 5C), MCF7 (mean 67% vs. 46%, Figure 5D), and T47D (mean 66% vs. 40%, Figure 5E). Overall, our protocol increased Vδ2 T cell-mediated cancer cell killing by 27% over ALN+IL-2 with the highest difference in killing efficiency observed for the lung cell line H1299 (Figure 5F).

To determine whether differences in HLA-E expression among the cancer cell lines contributed to the variable killing efficiency, we measured HLA-E expression in the different cell lines before and after the coculture with expanded Vδ2 T cells. Our results showed an increase in HLA-E expression upon coculture with Vδ2 T cells expanded with ALN+IL-2 but not upon coculture with IL-12 and CD137L supplementation (Figure S6). Therefore, our optimized protocol did not induce upregulation of HLA-E upon coculture and may have contributed to the enhanced killing of H1299 cells. Further investigation is needed to identify the mechanism of HLA-E upregulation and its overall implication in Vδ2 T cell-mediated cytotoxicity.

Late cytotoxicity

Next, we compared the cytotoxicity of expanded Vδ2 T cells from eight different donors against the same cell lines after overnight

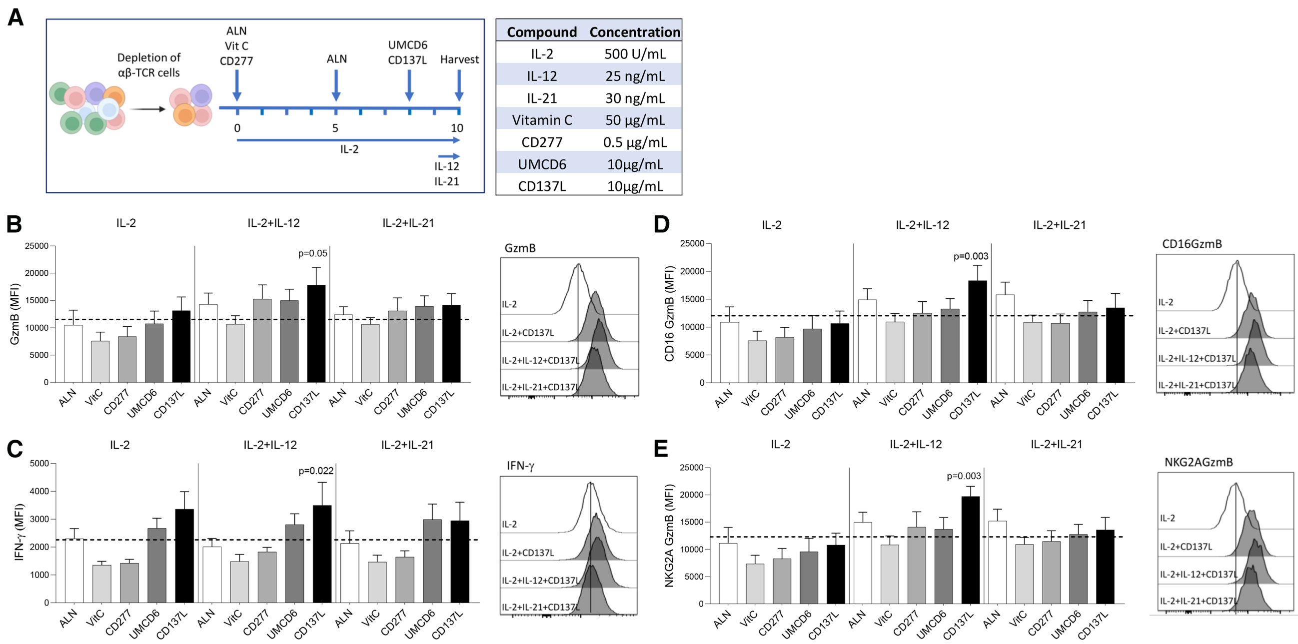

Figure 3. Comparison of Vδ2 T cell frequency and cytotoxic markers using different protocols in 10-day culture expansions

(A) Scheme of the culture conditions tested and concentrations of the different compounds. Frequency and representative histograms from four conditions of cytotoxic Vδ2 T cells expressing (B) granzyme B and (C) IFN-γ. Frequency of Vδ2 T cells producing GzmB and expressing (D) CD16 or (E) NKG2A. Horizontal dash lines represent the mean of all combination experiments. Representative histograms show the comparison between ALN+IL-2 and conditions receiving CD137L. *p* values represent the highest statistical difference among the different tested conditions (ANOVA).

coculture. Cells were cocultured at a 1:1 ratio, and the frequency of alive cancer cells was analyzed by flow cytometry (Figure 6A). Our results showed that Vδ2 T cells generated with ALN+IL-2 compared to our protocol had reduced overnight cytotoxicity against each target cell line including K562 (mean 26% vs. 47%), A2780 (mean 19% vs. 56%), H1299 (mean 39% vs. 77%), MCF7 (mean 50% vs. 77%), and T47D (mean 31% vs. 53%). Similar to the cytotoxicity assay, the frequency of alive cancer cells translated into an overall increased killing difference of 27% (Figure 6B). To assess whether the cytotoxic phenotype of expanded Vδ2 T cells was maintained after the coculture (Figures 4E–4I), we analyzed the phenotype of Vδ2 T cells after the overnight coculture in five donors. Mean GzmB production by Vδ2 T cells expanded with IL-12 and CD137L was higher against all cell lines compared to ALN+IL-2 (Figure 6C). After overnight coculture, both cytotoxic Vδ2 T cell populations $CD16^+$ (Figure 6D) and $NKG2A^+$ (Figure 6E) maintained increased GzmB production despite CD16 expression generally being higher in Vδ2 T cells expanded with ALN+IL-2 (Figure 6F). NKG2A expression was higher in cells expanded with IL-12 and CD137L (Figure 6G).

Finally, supernatants from the cocultures were harvested, and cytokines were measured by a multiplex assay that allowed simultaneous quantification of 12 analytes. The production of Fas and FasL, GzmA, and granulysin was similar between the two Vδ2 expansion protocols across the different cancer cells, except for the lung cell line H1299 that showed higher granulysin levels (Figure 6H). On the contrary, a consistently increased secretion of IFN-γ, TNF-α, and GzmB was observed for all the cell lines in cocultures of Vδ2 T cells expanded with IL-12 and CD137L, except for the breast cell line MCF7 that showed comparable IFN-γ production (Figure 6H). In summary, our results demonstrate that Vδ2 T cells expanded with ALN, CD137L, and IL-2/IL-12 supplementation increased tumor killing up to 40% in tumors of different origins.

DISCUSSION

Allogeneic Vδ2 T cell immunotherapy has provided evidence of benefit in recent clinical trials.[40,41,74,75] However, the lack of simplified methods to expand highly cytotoxic Vδ2 T cells has precluded such studies to take advantage of the full potential of this versatile cell population. Here, we have comprehensively tested strategic combinations of cytokines and compounds to develop a protocol that expands highly cytotoxic Vδ2 T cells with enhanced antitumor functions against cancer cells of diverse origin. Our protocol produced a final product with a mean purity above 94% and higher cytotoxic profile and function against several cancer cell lines. Vδ2 T cell expansion is achieved *in vitro* by exposing PBMCs to N-BPs, with most protocols utilizing ZOL and a secondary signal from cytokines like IL-2 or IL-15.[42] These methods induced polyclonal expansion of Vδ2 T cell subpopulations, and cytotoxic populations may not be specifically enriched in the process.[34] We have developed a method to selectively expand highly cytotoxic Vδ2 T cells with superior purity and antitumor functions compared to conventional approaches.

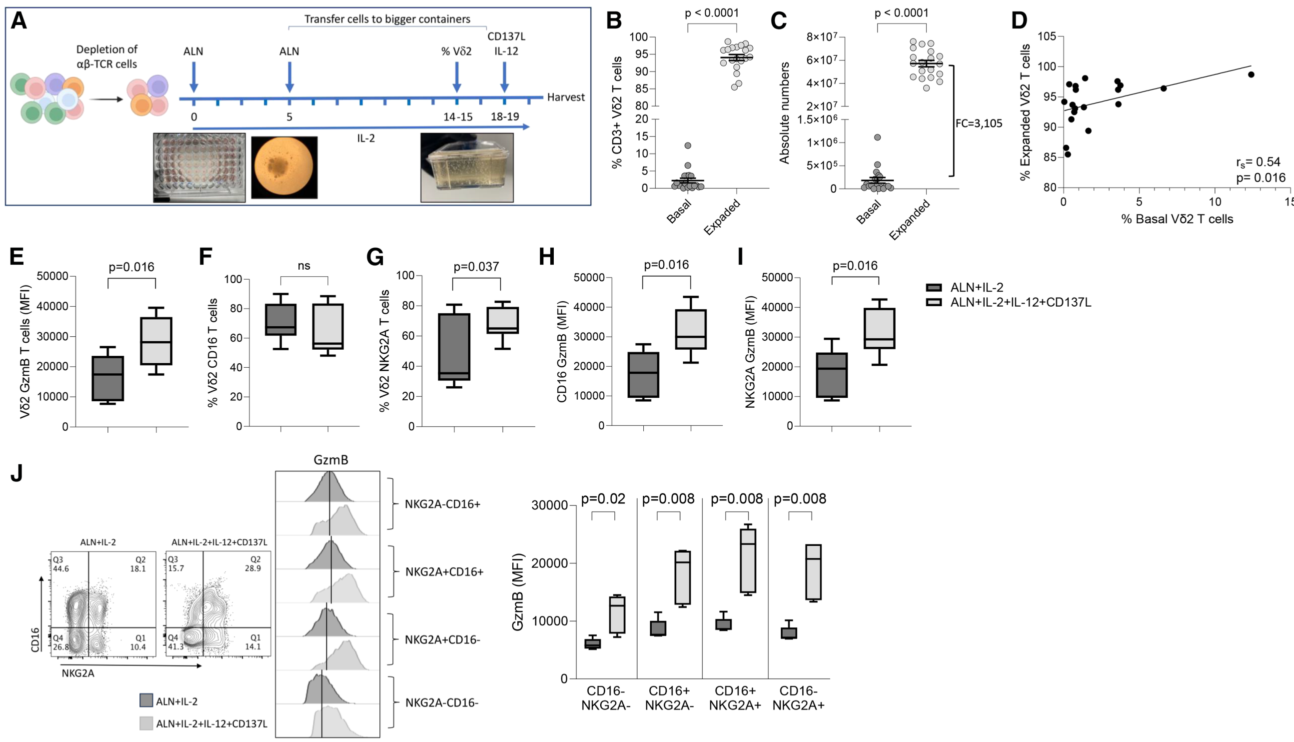

Figure 4. Increased frequency of cytotoxic Vδ2 T cells expanded with ALN, IL-2+IL-12 combined with CD137L

(A) Scheme of the expansion protocol. (B) Frequency and (C) absolute numbers of Vδ2 T cells before (basal) and after 18–19 days of expansion with ALN and IL-2. (D) Correlation between basal and expanded Vδ2 T cell frequencies. Spearman correlation test. Comparison of cytotoxic markers (E) GzmB and (F) CD16, (G) immune checkpoint NKG2A, and distribution of GzmB within (H) CD16$^+$ or (I) NKG2A$^+$ Vδ2$^+$ cells upon expansion with ALN+IL-2 compared to ALN+IL-2+IL-12+CD137L. (J) Distribution of GzmB production in CD16/NKG2A Vδ2 T cell populations. FC, fold change expansion. Wilcoxon matched-pairs signed-rank tests. ns, not significant.

Unexpectedly, we found that ALN was more potent inducing Vδ2 T cell expansion compared to ZOL or PAM. Additionally, a second pulse of ALN increased both the frequency and the viability at the end of the culture. αβ-TCR$^+$ cells are not required to expand Vδ2 T cells.[62] In addition, a previous study that used genetically modified artificial APCs showed that depletion of αβ T cells after 7 days in culture with ZOL and IL-2 improved the frequency of expanded Vδ2 T cells.[60,63,76] In our study, we depleted PBMCs of αβ T cells before starting the expansion, and cells were sequentially exposed to ALN and IL-2. This simple step induced the highest median Vδ2 T cell frequency while enhancing the purity of the final product to a mean of 94%. This significant improvement compared to other protocols[60,62,63,77] avoids the requirement of further purification steps. As expected, depletion of αβ-TCR$^+$ T cells did not modify the cytotoxic profile of expanded Vδ2 T cells. Our results showed that the majority of expanded Vδ2 T cells were predominantly effector memory, in agreement with other studies that showed an expansion of memory phenotypes.[48,52,78]

In addition to IL-2 or IL-15, other cytokines have been used to improve the overall Vδ2 T cell expansion, viability, or cytotoxicity, including IL-12, IL-18, or IL-21, and other compounds such as transforming growth factor β.[48,49,59,65,78–81] Vitamin C is an essential vitamin with a critical impact on immune cells, and Kabelitz's group demonstrated its beneficial effects on the proliferation and effector function of expanded Vδ2 T cells as it may act as an epigenetic modifier leading to increased cytokine production and metabolic activity.[55,82] Our results confirmed that vitamin C induced the highest frequency of Vδ2 T cells, viability, and CD16 and CD56 expression. However, these populations were not enriched in GzmB or IFN-γ, suggesting that higher doses may be required to induce an increase in GzmB production, as reported for CD8$^+$ T cells.[83] In addition, our method included vitamin C for the entire expansion process with N-BPs, while, in previous studies, it was added toward the end of the 14-period culture and prior to re-stimulation with a pyrophosphate.[55] Similarly, targeting the CD137-CD137L axis has been shown to improve the cytotoxic capacity of effector cells including Vδ2, NK, and CD8 T cells[71,72,84,85] and constitutes an additional strategy to enhance anticancer immune responses.[86] In our study, the addition of CD137L induced the accumulation of GzmB and IFN-γ. Interestingly, CD137L induced the expansion of double-positive CD16$^+$ NKG2A$^+$ Vδ2 T populations that were enriched in GzmB, compared to a protocol that entailed the use of ALN+IL-2 only.

The purity of our final cell product after a mean culture of 19 days was 94.1% with an average 3,105-fold expansion. In a recent study,

Table 2. Frequency, absolute numbers, fold change expansion, and expansion time for the different expansions

	Basal			Expanded				
Donor	Total cells	Vδ2 (%)	Absolute numbers	Total cells	Vδ2 (%)	Absolute numbers	Fold change	Time (days)
12	15,000,000	1.42	213,000	63,000,000	98.10	61,803,000	290.16	17
18	10,000,000	3.62	362,000	72,000,000	93.80	67,536,000	186.57	17
22	9,000,000	12.40	1,116,000	75,000,000	98.70	74,025,000	66.34	17
23	13,000,000	0.80	104,000	65,000,000	96.20	62,530,000	601.25	17
24	9,000,000	0.79	71,100	77,000,000	96.80	74,536,000	1,048.33	19
25	7,400,000	0.37	27,380	59,000,000	97.10	57,289,000	2,092.37	20
26	7,400,000	0.05	3,478	55,000,000	94.20	51,810,000	14,896.50	20
27	8,000,000	3.60	288,000	43,000,000	96.20	41,366,000	143.64	20
28	16,000,000	1.34	214,400	68,000,000	93.30	63,444,000	295.92	20
29	14,000,000	3.75	525,000	73,000,000	96.90	70,737,000	134.74	20
30	7,300,000	3.56	259,880	52,000,000	97.60	50,752,000	195.30	20
31	9,100,000	0.28	25,480	68,000,000	85.50	58,140,000	2,281.79	19
32	10,300,000	0.16	16,480	88,000,000	86.60	76,208,000	4,624.28	19
33	8,500,000	0.70	59,500	52,000,000	92.50	48,100,000	808.41	19
34	10,200,000	0.75	76,500	61,000,000	93.10	56,791,000	742.37	19
35	9,300,000	0.59	54,870	48,000,000	93.70	44,976,000	819.69	18
36	196,800	1.64	3,228	40,230,000	89.40	35,965,620	11,143.43	18
37	484,500	0.51	2,471	48,230,000	91.30	44,033,990	17,820.68	18
38	859,300	6.61	56,800	47,236,000	96.40	45,535,504	801.69	18
Mean	8,686,347	2.26	183,135	60,773,474	94.07	57,135,690	3,104.92	19.00

ZOL and IL-2 in combination with genetically engineered K-562 artificial APCs achieved a 633-fold expansion after 10 days of culture.[63] Additional studies using the same protocol in Good Manufacturing Practices developed a process that could potentially expand Vδ2 T cells by more than 200,700-fold with a purity of 99% and a viability of 95%.[60]

The overall absolute Vδ2 T cell frequency was donor dependent with multiple factors impacting on such differences including the basal frequency of Vδ2 T cells, which in turn can depend on sex, age, and race, as we and others previously reported.[34,87,88] However, one limitation in our study is that the expansions were finalized depending on the number of cells required for downstream assays, and therefore, our fold change expansion calculation also included the time on culture as a variable. In addition, donor variability may also impact the expansion efficiency shown in previous studies.[89] Nonetheless, the frequency of Vδ2 T cells after expansion was above 94%, and the viability was consistently above 95%. In agreement with other studies, the frequency of Vδ2 cells expressing NKG2A and NKG2D was also enhanced upon expansion.[18,19,21] A study by Cazzetta et al. showed that NKG2A identified a distinct highly cytotoxic Vδ2 T cell population against leukemia cell lines.[18] In addition, we and others previously reported that CD16 is also a marker of highly cytotoxic Vδ2 T cells against both cancer cell lines and HIV-infected targets.[34–37] In line with these results, our study showed that expansion with ALN+IL-2 supplemented with IL-12+CD137L increased the production of both NKG2A and CD16 cell populations that were enriched in GzmB. IL-12 and CD137L signaling pathways synergize, promoting IFN-γ production and cytotoxic activity of γδ T cells.[90] Our study provides additional insights into the beneficial synergistic effects of this combination, showing selective expansion of cytotoxic Vδ2 T cell populations enriched in GzmB and enhanced antitumor functions.

To confirm that the higher frequency of $CD16^+GzmB^+$ and $NKG2A^+GzmB^+$ Vδ2 T cells was associated with increased antitumor function, we tested early and late cytotoxicity against several cancer cell lines including leukemia and lung, ovarian, and breast carcinomas. Our results demonstrated increased killing of Vδ2 T cells expanded with ALN+IL-2+IL-12+CD137L against all the cell lines analyzed compared to ALN+IL-2 with an overall enhanced antitumor function. CD16 identifies Vδ2 T cells with a distinct phenotype characterized by increased cytolytic potential that resembles mature NK cells.[37] In our study, we confirm that CD16 identifies a highly cytotoxic Vδ2 T cell subpopulation[22,34,36,61,91] enriched in GzmB. Indeed, increased secretion of GzmB, along with additional cytolytic mediators, was also quantified in the supernatant following coculture of ALN+IL-2+IL-12+CD137L-expanded Vδ2 T cells and tumor cells.

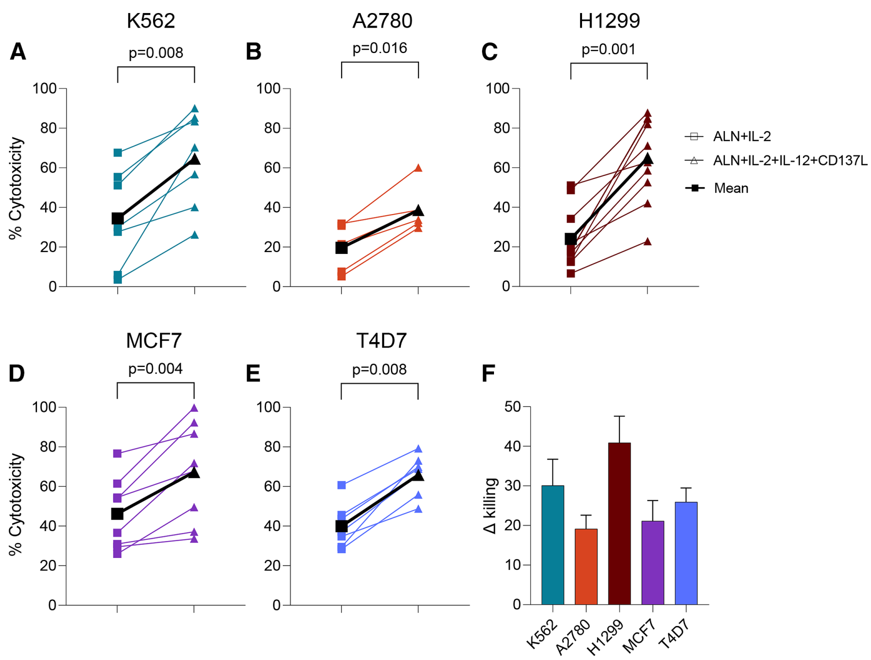

Figure 5. Increased cytolytic function of Vδ2 T cells expanded with an optimized protocol

Comparison of the cytotoxic function of Vδ2 T cells expanded with either ALN+IL-2 or ALN+IL-2+IL-12+CD137L in a non-radioactive cytotoxic assay, against cancer cell lines (A) K562, (B) H1299, (C) A2780, (D) MCF7, and (E) T47D. (F) Difference of cancer cell lines killing between Vδ2 T cells expanded with either protocol. Wilcoxon matched-pairs signed-rank tests. ns, not significant.

On the other hand, the involvement of HLA-E in tumor escape is a well-recognized phenomenon, and variable expression among different cancer cell lines may result in differences in killing efficacy.[18,26,92] HLA-E expression in H1299 cells increased upon coculture of ALN+IL-2-expanded Vδ2 T cells but not when the culture was supplemented with IL-12 and CD137L. Therefore, increased killing without inducing tumor resistance may explain why the biggest difference in Vδ2 T cell killing between protocols was with the cell line H1299.

Despite the recent exciting results in treating patients with late-stage lung and liver cancer,[41] one of the challenges faced by current allogenic Vδ2 T cell immunotherapy is the lack of a manageable protocol that expands discrete cytolytic phenotypes.[12,42,43] We present a straightforward method for expanding highly purified Vδ2 T cells, enhancing their cytolytic and cytotoxic activity against tumors. This approach shows promise for improving antitumor function and warrants translational implementation.

MATERIALS AND METHODS

Samples, cell lines, and compounds

Buffy coats were obtained from healthy volunteer donors from the Gulf Coast Regional Blood Center (Texas).

The lymphoblast K562 cell line, obtained from the American Type Culture Collection (ATCC), was derived from the bone marrow of a patient with chronic myelogenous leukemia. The lung cancer cell line H1299 is an epithelial carcinoma cell line, and A2780 is an ovarian cancer cell line from an endometroid adenocarcinoma (both kindly provided by Dr. Chiappinelli). MCF7 is an epithelial cell line derived from breast adenocarcinoma, and T47D epithelial cells were isolated from a patient with breast carcinoma (kindly provided by Dr. Bosque).

N-BPs PAM, ZOL, and ALN (MilliporeSigma) were tested at different concentrations (Table 1). Vitamin C (2-phospho-L-ascorbic acid trisodium salt, MilliporeSigma) was used at 50 μg/mL, CD277 (clone 20.1, BT3.1, Thermo Fisher Scientific) at 0.5 μg/mL, CD6 (clone UMCD6, MilliporeSigma) at 10 μg/mL, and CD137L (AdipoGen Life Sciences) at 10 μg/mL. IL-2 was used at 500 U/mL, IL-12 at 25 ng/mL, and IL-21 at 30 ng/mL (PeproTech) (Table 2). Concentrations were selected based on previous studies.[6,55,70,71]

Expansion of Vδ2 T cells

PBMCs were isolated from buffy coats by Ficoll gradient centrifugation, cultured at 1×10^6 cells/mL in complete RPMI (cRPMI) that consisted of RPMI-1640 medium supplemented with 10% fetal bovine serum (FBS) and 1% Penicillin/Streptomycin in the presence of indicated concentrations of N-BPs and 500 U/mL IL-2. On day 5 of culture, a second dose of N-BPs was added, and media were replaced every 2–3 days until day 10 of culture. In some experiments, αβ-TCR$^+$ T cells were depleted using the EasySep human TCR αβ Depletion Kit (STEMCELL Technologies) prior to the expansion. In addition, we compared the effect of the combination of ALN with IL-2 or IL-2+IL-12 or IL-2+IL-21 in the presence or absence of UMCD6, CD137L, vitamin C, or CD277.

Phenotypic analysis of expanded Vδ2 T cells by flow cytometry

At the end of the culture period, cells were harvested, counted three times using the Muse cell analyzer (Cytek Biosciences), washed twice, and stained with a viability dye (Zombie Aqua Fixable viability kit, BioLegend) for 15 min at room temperature. Cells were then washed and resuspended in staining buffer (PBS + 2% FBS), followed by surface staining with CD3 (clone SK7), Vδ2 (clone B6), CD16 (clone 3G8), CD56 (clone 5.1H11), NKG2D (clone 1D11), NKG2A (clone cS19004C), CD95 (clone DX2), HLA-E (clone 3D12), CD28 (clone CD28.2), and CD27 (clone O323) (antibodies were all purchased from BioLegend). Cells were incubated for 20 min on ice in the dark, washed, and fixed in 2% paraformaldehyde before

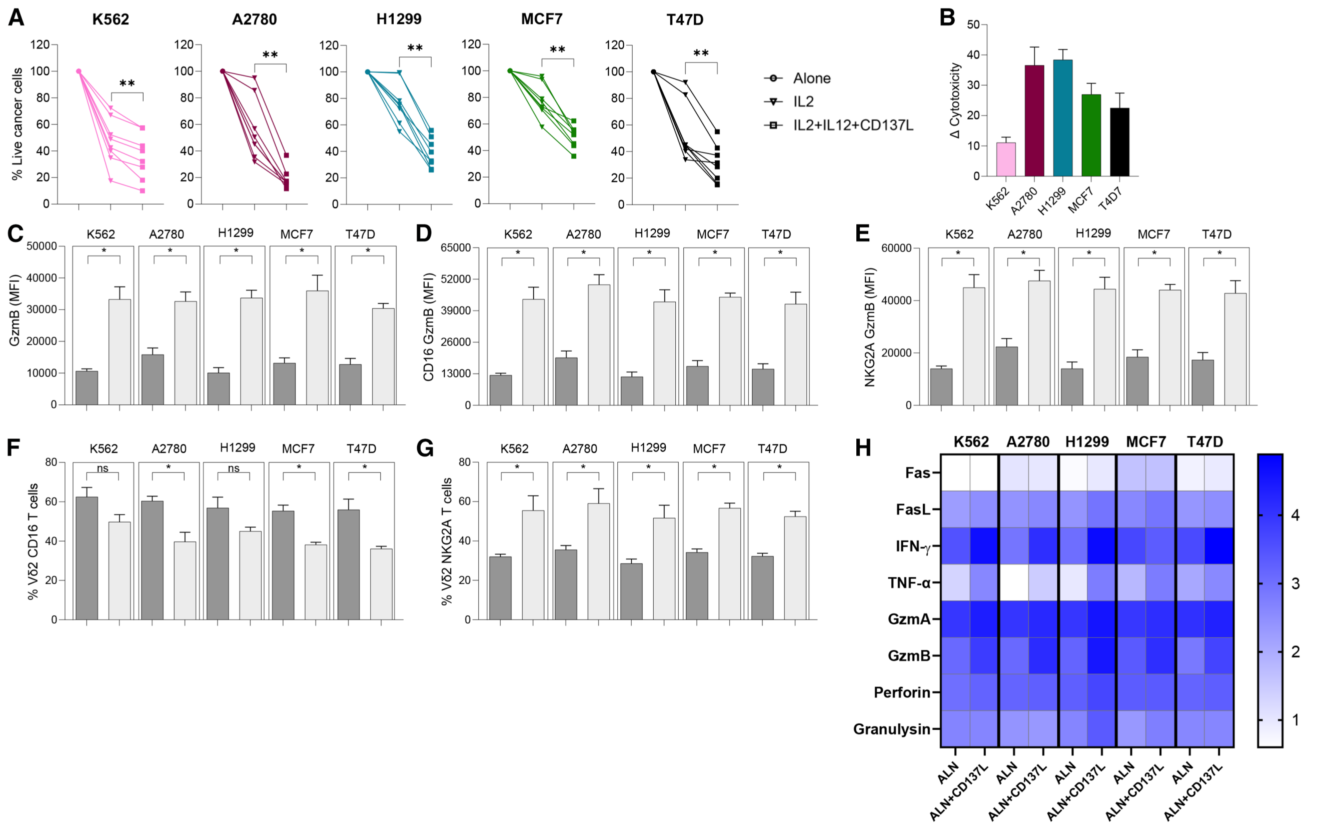

Figure 6. Increased Vδ2 T cell cytolytic function against cancer cell lines in overnight cocultures

(A) Comparison of cytotoxic function by flow cytometry against cancer cell lines K562, H1299, A2780, MCF7, and T47D and (B) difference in cytotoxicity of Vδ2 T cells from eight donors expanded with either ALN+IL-2 or ALN+IL-2+IL-12+CD137L. Intracellular production of GzmB in (C) total (D) CD16^{+} and (E) NKG2A^{+} Vδ2 T cells and frequency of (F) CD16 and (G) NKG2A expression upon coculture of different cancer cell lines with Vδ2 T cells expanded with ALN+IL-2 (dark gray) or ALN+IL-2+IL-12+CD137L (light gray). Wilcoxon matched-pairs signed-rank tests. ns, not significant. Circles represent cancer cell lines cultured alone, and cocultures with Vδ2 T cells expanded with IL-2 are shown with triangles and those expanded with IL-2+IL-12+CD137L with squares. (H) Production of cytolytic molecules in the supernatant upon coculture.

acquisition. For intracellular staining, after washing the surface staining, cells were fixed and permeabilized with Cytofix/Cytoperm (BD Biosciences) for 30 min on ice in the dark followed by intracellular staining with GzmB (clone GB11) and IFN-γ (clone B27). After 20 min of incubation on ice in the dark, cells were washed twice, acquired on a BD LSR Fortessa TM X-20 instrument (BD), and analyzed using FlowJo (FlowJo v.10.8.1).

Coculture of cancer cell lines with expanded Vδ2 T cells

Cancer cells were cultured in cRPMI following the ATCC culture recommendations. Vδ2 T cells were expanded following the indicated protocol until they reached >90% of the total cells in culture when they were harvested and washed. Vδ2 T cells were cocultured with the cancer cell lines at a 1:1 effector-to-target (E:T) ratio, as indicated in the results. Cytotoxicity was measured either by flow cytometry analyzing the viability of the cancer cells or by a dissociation-enhanced lanthanide fluorescence immunoassay (DELFIA) cell cytotoxicity assay, as described in the following section.

DELFIA cell cytotoxic assay

The DELFIA cell cytotoxicity assay was performed to measure the cytolytic capacity of Vδ2 T cells expanded with the indicated protocol against cancer cell lines A2780, H1299, K562, MCF7, and T4D7 according to the manufacturer's protocol (PerkinElmer). Adherent cell lines A2780, H1299, MCF7, and T4D7 were washed 5 times, and K562 cells were washed 4 times after labeling with BATDA (1,2-bis(2-aminophenoxy) ethane-N,N, N9,N9-tetraacetic acid). Then, Vδ2 T cells were incubated with the different cancer cell lines in cRPMI at a 1:1 E:T ratio. Supernatants were then harvested and incubated with Europium solution in OptiPlate 96-well microplate for 15 min. The plates were read in a FLUOstar Omega instrument (BMG Labtech). The percentage of specific killing was calculated with the following formula: 100 × [(BATDA release wit Vδ2 T cells) − (BATDA release with no Vδ2 T cells)]/[(maximum BATDA release) − (BATDA release with no Vδ2 T cells)]. As controls, the supernatant from the last wash of the target cells was used as background, target cells as spontaneous, and target cells with lysis buffer as release.

Multiplex cytokine quantification

Supernatants from cocultures of expanded Vδ2 T cells and cancer cell lines were harvested after 18 h and frozen at −80°C for later analyses. The samples were thawed to room temperature and then cytokines and cytotoxic soluble factors were analyzed using the LEGENDplex Human CD8/NK Panel 13-plex V02 (BioLegend, 741187). The assay was conducted per the manufacturer's protocol with samples run in duplicate. Samples were acquired using the BD LSRFortessa X-20 (BD Biosciences) and analyzed using the LEGENDplex Data Analysis Software Suite v.2023-02-15 (BioLegend/Qognit). Values were log-transformed prior to heatmap visualization in GraphPad PRISM v.10.2.

Statistical analysis

Non-parametric Mann-Whitney U tests or Wilcoxon matched-pairs signed-rank tests were used to compare differences between groups or conditions, and Spearman tests were used for correlative analysis. The effect of N-BP concentration on Vδ2 T cell frequency and cytotoxic markers was examined through plots of means and medians, which were used to estimate optimal concentrations. Cell frequencies at these optimal concentrations were then compared across the 15 combination treatments. For each type of cell frequency or marker, differences among the 15 treatments were tested through repeated measures analysis of variance contrasts with the Chi-Muller Epsilon adjustment for deviation from the compound symmetry assumption. To avoid an excess type 1 error rate, each contrast compared each combination to the mean of the remaining 14 combinations.

DATA AVAILABILITY

The corresponding author will make data available for researchers to interpret, verify, and extend the research upon request.

ACKNOWLEDGMENTS

This research was funded by the National Institute of Allergy and Infectious Diseases (NIAID) of the National Institutes of Health (NIH) under award nos. R01-AI125097 and R21-AI157864 to N.S.-S. and facilitated by the services provided by the District of Columbia Center for AIDS Research under an NIH-funded program (P30AI117970).

AUTHOR CONTRIBUTIONS

M.S. performed research, analyzed the data, and wrote the manuscript. B.T.M. performed research and edited the manuscript. E.K.M. performed research. S.S. performed statistical analyses of the data. A.B. reviewed the manuscript. N.S.-S. designed and supervised the study and wrote and edited the manuscript. All authors read and approved the final version of the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.omton.2025.200996.

REFERENCES

1. Morita, C.T., Jin, C., Sarikonda, G., and Wang, H. (2007). Nonpeptide antigens, presentation mechanisms, and immunological memory of human Vgamma2Vdelta2 T cells: discriminating friend from foe through the recognition of prenyl pyrophosphate antigens. Immunol. Rev. *215*, 59–76.
2. Adams, E.J., Gu, S., and Luoma, A.M. (2015). Human gamma delta T cells: Evolution and ligand recognition. Cell. Immunol. *296*, 31–40.
3. Green, A.E., Lissina, A., Hutchinson, S.L., Hewitt, R.E., Temple, B., James, D., Boulter, J.M., Price, D.A., and Sewell, A.K. (2004). Recognition of nonpeptide antigens by human V gamma 9V delta 2 T cells requires contact with cells of human origin. Clin. Exp. Immunol. *136*, 472–482.
4. Jomaa, H., Feurle, J., Lühs, K., Kunzmann, V., Tony, H.P., Herderich, M., and Wilhelm, M. (1999). Vgamma9/Vdelta2 T cell activation induced by bacterial low molecular mass compounds depends on the 1-deoxy-D-xylulose 5-phosphate pathway of isoprenoid biosynthesis. FEMS Immunol. Med. Microbiol. *25*, 371–378.
5. Harly, C., Guillaume, Y., Nedellec, S., Peigné, C.M., Mönkkönen, H., Mönkkönen, J., Li, J., Kuball, J., Adams, E.J., Netzer, S., et al. (2012). Key implication of CD277/butyrophilin-3 (BTN3A) in cellular stress sensing by a major human γδ T-cell subset. Blood *120*, 2269–2279.
6. Rigau, M., Ostrouska, S., Fulford, T.S., Johnson, D.N., Woods, K., Ruan, Z., McWilliam, H.E.G., Hudson, C., Tutuka, C., Wheatley, A.K., et al. (2020). Butyrophilin 2A1 is essential for phosphoantigen reactivity by γδ T cells. Science *367*, eaay5516.
7. Vavassori, S., Kumar, A., Wan, G.S., Ramanjaneyulu, G.S., Cavallari, M., El Daker, S., Beddoe, T., Theodossis, A., Williams, N.K., Gostick, E., et al. (2013). Butyrophilin 3A1 binds phosphorylated antigens and stimulates human γδ T cells. Nat. Immunol. *14*, 908–916.
8. Tanaka, Y., Morita, C.T., Tanaka, Y., Nieves, E., Brenner, M.B., and Bloom, B.R. (1995). Natural and synthetic non-peptide antigens recognized by human gamma delta T cells. Nature *375*, 155–158.
9. Alexander, A.A.Z., Maniar, A., Cummings, J.S., Hebbeler, A.M., Schulze, D.H., Gastman, B.R., Pauza, C.D., Strome, S.E., and Chapoval, A.I. (2008). Isopentenyl pyrophosphate-activated CD56+ {gamma}{delta} T lymphocytes display potent antitumor activity toward human squamous cell carcinoma. Clin. Cancer Res. *14*, 4232–4240.
10. Yuan, L., Ma, X., Yang, Y., Qu, Y., Li, X., Zhu, X., Ma, W., Duan, J., Xue, J., Yang, H., et al. (2023). Phosphoantigens glue butyrophilin 3A1 and 2A1 to activate Vγ9Vδ2 T cells. Nature *621*, 840–848.
11. Silva-Santos, B., Serre, K., and Norell, H. (2015). γδ T cells in cancer. Nat. Rev. Immunol. *15*, 683–691.
12. Kabelitz, D., Serrano, R., Kouakanou, L., Peters, C., and Kalyan, S. (2020). Cancer immunotherapy with γδ T cells: many paths ahead of us. Cell. Mol. Immunol. *17*, 925–939.
13. Poonia, B., and Pauza, C.D. (2012). Gamma delta T cells from HIV+ donors can be expanded in vitro by zoledronate/interleukin-2 to become cytotoxic effectors for antibody-dependent cellular cytotoxicity. Cytotherapy *14*, 173–181.
14. Göbel, A., Thiele, S., Browne, A.J., Rauner, M., Zinna, V.M., Hofbauer, L.C., and Rachner, T.D. (2016). Combined inhibition of the mevalonate pathway with statins and zoledronic acid potentiates their anti-tumor effects in human breast cancer cells. Cancer Lett. *375*, 162–171.
15. Dalton, J.E., Howell, G., Pearson, J., Scott, P., and Carding, S.R. (2004). Fas-Fas ligand interactions are essential for the binding to and killing of activated macrophages by gamma delta T cells. J. Immunol. *173*, 3660–3667.
16. Ponomarev, E.D., and Dittel, B.N. (2005). Gamma delta T cells regulate the extent and duration of inflammation in the central nervous system by a Fas ligand-dependent mechanism. J. Immunol. *174*, 4678–4687.
17. Vantourout, P., and Hayday, A. (2013). Six-of-the-best: unique contributions of γδ T cells to immunology. Nat. Rev. Immunol. *13*, 88–100.
18. Cazzetta, V., Bruni, E., Terzoli, S., Carenza, C., Franzese, S., Piazza, R., Marzano, P., Donadon, M., Torzilli, G., Cimino, M., et al. (2021). NKG2A expression identifies a subset of human Vδ2 T cells exerting the highest antitumor effector functions. Cell Rep. *37*, 109871.
19. Rincon-Orozco, B., Kunzmann, V., Wrobel, P., Kabelitz, D., Steinle, A., and Herrmann, T. (2005). Activation of V gamma 9V delta 2 T cells by NKG2D. J. Immunol. *175*, 2144–2151.
20. Peters, C., Meyer, A., Kouakanou, L., Feder, J., Schricker, T., Lettau, M., Janssen, O., Wesch, D., and Kabelitz, D. (2019). TGF-β enhances the cytotoxic activity of Vδ2 T cells. OncoImmunology *8*, e1522471.

21. Vermijlen, D., Ellis, P., Langford, C., Klein, A., Engel, R., Willimann, K., Jomaa, H., Hayday, A.C., and Eberl, M. (2007). Distinct cytokine-driven responses of activated blood gammadelta T cells: insights into unconventional T cell pleiotropy. J. Immunol. *178*, 4304–4314.

22. Lee, D., Dunn, Z.S., Guo, W., Rosenthal, C.J., Penn, N.E., Yu, Y., Zhou, K., Li, Z., Ma, F., Li, M., et al. (2023). Unlocking the potential of allogeneic Vδ2 T cells for ovarian cancer therapy through CD16 biomarker selection and CAR/IL-15 engineering. Nat. Commun. *14*, 6942.

23. Mingari, M.C., Ponte, M., Bertone, S., Schiavetti, F., Vitale, C., Bellomo, R., Moretta, A., and Moretta, L. (1998). HLA class I-specific inhibitory receptors in human T lymphocytes: interleukin 15-induced expression of CD94/NKG2A in superantigen- or alloantigen-activated CD8+ T cells. Proc. Natl. Acad. Sci. USA *95*, 1172–1177.

24. Zaghi, E., Calvi, M., Marcenaro, E., Mavilio, D., and Di Vito, C. (2019). Targeting NKG2A to elucidate natural killer cell ontogenesis and to develop novel immune-therapeutic strategies in cancer therapy. J. Leukoc. Biol. *105*, 1243–1251.

25. Li, Y., Li, Z., Tang, Y., Zhuang, X., Feng, W., Boor, P.P.C., Buschow, S., Sprengers, D., and Zhou, G. (2024). Unlocking the therapeutic potential of the NKG2A-HLA-E immune checkpoint pathway in T cells and NK cells for cancer immunotherapy. J. Immunother. Cancer *12*, e009934.

26. Kaiser, B.K., Pizarro, J.C., Kerns, J., and Strong, R.K. (2008). Structural basis for NKG2A/CD94 recognition of HLA-E. Proc. Natl. Acad. Sci. USA *105*, 6696–6701.

27. Taveirne, S., Filtjens, J., Van Ammel, E., De Colvenaer, V., Kerre, T., Taghon, T., Vandekerckhove, B., Plum, J., Held, W., and Leclercq, G. (2011). Inhibitory receptors specific for MHC class I educate murine NK cells but not CD8αα intestinal intraepithelial T lymphocytes. Blood *118*, 339–347.

28. Abd Hamid, M., Wang, R.Z., Yao, X., Fan, P., Li, X., Chang, X.M., Feng, Y., Jones, S., Maldonado-Perez, D., Waugh, C., et al. (2019). Enriched HLA-E and CD94/NKG2A Interaction Limits Antitumor CD8(+) Tumor-Infiltrating T Lymphocyte Responses. Cancer Immunol. Res. *7*, 1293–1306.

29. Gooden, M., Lampen, M., Jordanova, E.S., Leffers, N., Trimbos, J.B., van der Burg, S. H., Nijman, H., and van Hall, T. (2011). HLA-E expression by gynecological cancers restrains tumor-infiltrating $CD8^+$ T lymphocytes. Proc. Natl. Acad. Sci. USA *108*, 10656–10661.

30. Fisher, J.G., Doyle, A.D.P., Graham, L.V., Khakoo, S.I., and Blunt, M.D. (2022). Disruption of the NKG2A:HLA-E Immune Checkpoint Axis to Enhance NK Cell Activation against Cancer. Vaccines (Basel) *10*, 1993.

31. Ochoa, M.C., Minute, L., Rodriguez, I., Garasa, S., Perez-Ruiz, E., Inogés, S., Melero, I., and Berraondo, P. (2017). Antibody-dependent cell cytotoxicity: immunotherapy strategies enhancing effector NK cells. Immunol. Cell Biol. *95*, 347–355.

32. Tokuyama, H., Hagi, T., Mattarollo, S.R., Morley, J., Wang, Q., So, H.F., Moriyasu, F., Nieda, M., and Nicol, A.J. (2008). V gamma 9 V delta 2 T cell cytotoxicity against tumor cells is enhanced by monoclonal antibody drugs–rituximab and trastuzumab. Int. J. Cancer *122*, 2526–2534.

33. Braakman, E., van de Winkel, J.G., van Krimpen, B.A., Jansze, M., and Bolhuis, R.L. (1992). CD16 on human gamma delta T lymphocytes: expression, function, and specificity for mouse IgG isotypes. Cell. Immunol. *143*, 97–107.

34. Clohosey, M.L., Mann, B.T., Ryan, P.L., Apanasovich, T.V., Maggirwar, S.B., Pennington, D.J., and Soriano-Sarabia, N. (2020). Comparable Vδ2 Cell Functional Characteristics in Virally Suppressed People Living with HIV and Uninfected Individuals. Cells *9*, 2568.

35. James, K.S., Trumble, I., Clohosey, M.L., Moeser, M., Roan, N.R., Adimora, A.A., Joseph, S.B., Archin, N.M., Hudgens, M., and Soriano-Sarabia, N. (2019). Measuring the contribution of gammadelta T cells to the persistent HIV reservoir. Aids *34*, 363–371.

36. Garrido, C., Clohosey, M.L., Whitworth, C.P., Hudgens, M., Margolis, D.M., and Soriano-Sarabia, N. (2018). gammadelta T cells: an immunotherapeutic approach for HIV cure strategies. JCI Insight *3*, e120121.

37. Farrington, L.A., Callaway, P.C., Vance, H.M., Baskevitch, K., Lutz, E., Warrier, L., McIntyre, T.I., Budker, R., Jagannathan, P., Nankya, F., et al. (2020). Opsonized antigen activates Vδ2+ T cells via CD16/FCγRIIIa in individuals with chronic malaria exposure. PLoS Pathog. *16*, e1008997.

38. Wilhelm, M., Smetak, M., Schaefer-Eckart, K., Kimmel, B., Birkmann, J., Einsele, H., and Kunzmann, V. (2014). Successful adoptive transfer and in vivo expansion of haploidentical γδ T cells. J. Transl. Med. *12*, 45.

39. Alnaggar, M., Xu, Y., Li, J., He, J., Chen, J., Li, M., Wu, Q., Lin, L., Liang, Y., Wang, X., et al. (2019). Allogenic Vγ9Vδ2 T cell as new potential immunotherapy drug for solid tumor: a case study for cholangiocarcinoma. J. Immunother. Cancer *7*, 36.

40. Zhang, T., Chen, J., Niu, L., Liu, Y., Ye, G., Jiang, M., and Qi, Z. (2022). Clinical Safety and Efficacy of Locoregional Therapy Combined with Adoptive Transfer of Allogeneic γδ T Cells for Advanced Hepatocellular Carcinoma and Intrahepatic Cholangiocarcinoma. J. Vasc. Interv. Radiol. *33*, 19–27.e3.

41. Xu, Y., Xiang, Z., Alnaggar, M., Kouakanou, L., Li, J., He, J., Yang, J., Hu, Y., Chen, Y., Lin, L., et al. (2021). Allogeneic Vγ9Vδ2 T-cell immunotherapy exhibits promising clinical safety and prolongs the survival of patients with late-stage lung or liver cancer. Cell. Mol. Immunol. *18*, 427–439.

42. Sanz, M., Mann, B.T., Chitrakar, A., and Soriano-Sarabia, N. (2022). Human Vδ2 T Cells and Their Versatility for Immunotherapeutic Approaches. Cells *11*, 3572.

43. Ryan, P.L., Sumaria, N., Holland, C.J., Bradford, C.M., Izotova, N., Grandjean, C.L., Jawad, A.S., Bergmeier, L.A., and Pennington, D.J. (2016). Heterogeneous yet stable Vdelta2(+) T-cell profiles define distinct cytotoxic effector potentials in healthy human individuals. Proc. Natl. Acad. Sci. USA *113*, 14378–14383.

44. Mann, B.T., Sambrano, E., 3rd, Maggirwar, S.B., and Soriano-Sarabia, N. (2020). Boosting the Immune System for HIV Cure: A γδ T Cell Perspective. Front. Cell. Infect. Microbiol. *10*, 221.

45. Eberl, M., Engel, R., Beck, E., and Jomaa, H. (2002). Differentiation of human gamma-delta T cells towards distinct memory phenotypes. Cell. Immunol. *218*, 1–6.

46. Thedrez, A., Harly, C., Morice, A., Salot, S., Bonneville, M., and Scotet, E. (2009). IL-21-mediated potentiation of antitumor cytolytic and proinflammatory responses of human V gamma 9V delta 2 T cells for adoptive immunotherapy. J. Immunol. *182*, 3423–3431.

47. Kunzmann, V., Bauer, E., and Wilhelm, M. (1999). Gamma/delta T-cell stimulation by pamidronate. N. Engl. J. Med. *340*, 737–738.

48. Van Acker, H.H., Anguille, S., Willemen, Y., Van den Bergh, J.M., Berneman, Z.N., Lion, E., Smits, E.L., and Van Tendeloo, V.F. (2016). Interleukin-15 enhances the proliferation, stimulatory phenotype, and antitumor effector functions of human gamma delta T cells. J. Hematol. Oncol. *9*, 101.

49. Capsomidis, A., Benthall, G., Van Acker, H.H., Fisher, J., Kramer, A.M., Abeln, Z., Majani, Y., Gileadi, T., Wallace, R., Gustafsson, K., et al. (2018). Chimeric Antigen Receptor-Engineered Human Gamma Delta T Cells: Enhanced Cytotoxicity with Retention of Cross Presentation. Mol. Ther. *26*, 354–365.

50. Schilbach, K., Welker, C., Krickeberg, N., Kaißer, C., Schleicher, S., and Hashimoto, H. (2020). In the Absence of a TCR Signal IL-2/IL-12/18-Stimulated γδ T Cells Demonstrate Potent Anti-Tumoral Function Through Direct Killing and Senescence Induction in Cancer Cells. Cancers (Basel) *12*, 130.

51. Nakanishi, K., Yoshimoto, T., Tsutsui, H., and Okamura, H. (2001). Interleukin-18 regulates both Th1 and Th2 responses. Annu. Rev. Immunol. *19*, 423–474.

52. Nussbaumer, O., Gruenbacher, G., Gander, H., Komuczki, J., Rahm, A., and Thurnher, M. (2013). Essential requirements of zoledronate-induced cytokine and γδ T cell proliferative responses. J. Immunol. *191*, 1346–1355.

53. Bold, A., Gross, H., Holzmann, E., Knop, S., Hoeres, T., and Wilhelm, M. (2023). An optimized cultivation method for future in vivo application of γδ T cells. Front. Immunol. *14*, 1185564.

54. Assy, L., Khalil, S.M., Attia, M., and Salem, M.L. (2023). IL-12 conditioning of peripheral blood mononuclear cells from breast cancer patients promotes the zoledronate-induced expansion of γδ T cells in vitro and enhances their cytotoxic activity and cytokine production. Int. Immunopharmacol. *114*, 109402.

55. Kouakanou, L., Xu, Y., Peters, C., He, J., Wu, Y., Yin, Z., and Kabelitz, D. (2020). Vitamin C promotes the proliferation and effector functions of human γδ T cells. Cell. Mol. Immunol. *17*, 462–473.

56. Juraske, C., Krissmer, S.M., Teuber, E.S., Parigiani, M.A., Strietz, J., Wesch, D., Kabelitz, D., Minguet, S., and Schamel, W.W. (2024). Reprogramming of human γδ T cells by expression of an anti-CD19 TCR fusion construct (εTRuC) to enhance tumor killing. J. Leukoc. Biol. *115*, 293–305.

57. Hintz, M., Reichenberg, A., Altincicek, B., Bahr, U., Gschwind, R.M., Kollas, A.K., Beck, E., Wiesner, J., Eberl, M., and Jomaa, H. (2001). Identification of (E)-4-hydroxy-3-methyl-but-2-enyl pyrophosphate as a major activator for human gammadelta T cells in Escherichia coli. FEBS Lett. *509*, 317–322.

58. Domae, E., Hirai, Y., Ikeo, T., Goda, S., and Shimizu, Y. (2017). Cytokine-mediated activation of human ex vivo-expanded Vγ9Vδ2 T cells. Oncotarget *8*, 45928–45942.

59. Kondo, M., Izumi, T., Fujieda, N., Kondo, A., Morishita, T., Matsushita, H., and Kakimi, K. (2011). Expansion of human peripheral blood γδ T cells using zoledronate. J. Vis. Exp. 3182.

60. Landin, A.M., Cox, C., Yu, B., Bejanyan, N., Davila, M., and Kelley, L. (2021). Expansion and Enrichment of Gamma-Delta (γδ) T Cells from Apheresed Human Product. J. Vis. Exp. 175.

61. Sanz, M., Mann, B.T., Ryan, P.L., Bosque, A., Pennington, D.J., Hackstein, H., and Soriano-Sarabia, N. (2023). Deep characterization of human γδ T cell subsets defines shared and lineage-specific traits. Front. Immunol. *14*, 1148988.

62. Soriano-Sarabia, N., Sandvold, H., Jomaa, H., Kubin, T., Bein, G., and Hackstein, H. (2012). Primary MHC-class II(+) cells are necessary to promote resting Vδ2 cell expansion in response to (E)-4-hydroxy-3-methyl-but-2-enyl-pyrophosphate and isopentenyl pyrophosphate. J. Immunol. *189*, 5212–5222.

63. Boucher, J.C., Yu, B., Li, G., Shrestha, B., Sallman, D., Landin, A.M., Cox, C., Karyampudi, K., Anasetti, C., Davila, M.L., and Bejanyan, N. (2023). Large Scale Ex Vivo Expansion of γδ T cells Using Artificial Antigen-presenting Cells. J. Immunother. *46*, 5–13.

64. Fisher, J.P.H., Yan, M., Heuijerjans, J., Carter, L., Abolhassani, A., Frosch, J., Wallace, R., Flutter, B., Capsomidis, A., Hubank, M., et al. (2014). Neuroblastoma killing properties of Vδ2 and Vδ2-negative γδT cells following expansion by artificial antigen-presenting cells. Clin. Cancer Res. *20*, 5720–5732.

65. Khan, M.W.A., Otaibi, A.A., Sherwani, S., Alshammari, E.M., Al-Zahrani, S.A., Khan, W.A., Alsukaibi, A.K.D., Alouffi, S., and Khan, S.N. (2021). Optimization of methods for peripheral blood mononuclear cells isolation and expansion of human gamma delta T cells. Bioinformation *17*, 460–469.

66. Schilbach, K., Krickeberg, N., Kaißer, C., Mingram, S., Kind, J., Siegers, G.M., and Hashimoto, H. (2020). Suppressive activity of Vδ2(+) γδ T cells on αβ T cells is licensed by TCR signaling and correlates with signal strength. Cancer Immunol. Immunother. *69*, 593–610.

67. Duault, C., Franchini, D.M., Familliades, J., Cayrol, C., Roga, S., Girard, J.P., Fournié, J.J., and Poupot, M. (2016). TCRVγ9 γδ T Cell Response to IL-33: A CD4 T Cell-Dependent Mechanism. J. Immunol. *196*, 493–502.

68. Jonus, H.C., Burnham, R.E., Ho, A., Pilgrim, A.A., Shim, J., Doering, C.B., Spencer, H.T., and Goldsmith, K.C. (2022). Dissecting the cellular components of ex vivo γδ T cell expansions to optimize selection of potent cell therapy donors for neuroblastoma immunotherapy trials. Oncoimmunology *11*, 2057012.

69. Wang, H., Henry, O., Distefano, M.D., Wang, Y.C., Räikkönen, J., Mönkkönen, J., Tanaka, Y., and Morita, C.T. (2013). Butyrophilin 3A1 plays an essential role in prenyl pyrophosphate stimulation of human Vγ2Vδ2 T cells. J. Immunol. *191*, 1029–1042.

70. Ruth, J.H., Gurrea-Rubio, M., Athukorala, K.S., Rasmussen, S.M., Weber, D.P., Randon, P.M., Gedert, R.J., Lind, M.E., Amin, M.A., Campbell, P.L., et al. (2021). CD6 is a target for cancer immunotherapy. JCI Insight *6*, e145662.

71. Pei, Y., Xiang, Z., Wen, K., Tu, C.R., Wang, X., Zhang, Y., Mu, X., Liu, Y., and Tu, W. (2022). CD137 Costimulation Enhances the Antitumor Activity of Vγ9Vδ2-T Cells in IL-10-Mediated Immunosuppressive Tumor Microenvironment. Front. Immunol. *13*, 872122.

72. Pei, Y., Wen, K., Xiang, Z., Huang, C., Wang, X., Mu, X., Wen, L., Liu, Y., and Tu, W. (2020). CD137 costimulation enhances the antiviral activity of Vγ9Vδ2-T cells against influenza virus. Signal Transduct. Target. Ther. *5*, 74.

73. Meraviglia, S., and Dieli, F. (2021). Vitamin C as a promoter of γδ T cells. Cell. Mol. Immunol. *18*, 510–512.

74. Vydra, J., Cosimo, E., Lesný, P., Wanless, R.S., Anderson, J., Clark, A.G., Scott, A., Nicholson, E.K., and Leek, M. (2023). A Phase I Trial of Allogeneic γδ T Lymphocytes From Haploidentical Donors in Patients With Refractory or Relapsed Acute Myeloid Leukemia. Clin. Lymphoma Myeloma Leuk. *23*, e232–e239.

75. Bold, A., Gaertner, J., Bott, A., Mordstein, V., Schaefer-Eckart, K., and Wilhelm, M. (2023). Haploidentical γδ T Cells Induce Complete Remission in Chemorefractory B-cell Non-Hodgkin Lymphoma. J. Immunother. *46*, 56–58.

76. Deniger, D.C., Maiti, S.N., Mi, T., Switzer, K.C., Ramachandran, V., Hurton, L.V., Ang, S., Olivares, S., Rabinovich, B.A., Huls, M.H., et al. (2014). Activating and propagating polyclonal gamma delta T cells with broad specificity for malignancies. Clin. Cancer Res. *20*, 5708–5719.

77. Verkerk, T., Pappot, A.T., Jorritsma, T., King, L.A., Duurland, M.C., Spaapen, R.M., and van Ham, S.M. (2024). Isolation and expansion of pure and functional γδ T cells. Front. Immunol. *15*, 1336870.

78. Li, W., Kubo, S., Okuda, A., Yamamoto, H., Ueda, H., Tanaka, T., Nakamura, H., Yamanishi, H., Terada, N., and Okamura, H. (2010). Effect of IL-18 on expansion of gammadelta T cells stimulated by zoledronate and IL-2. J. Immunother. *33*, 287–296.

79. Mariani, S., Muraro, M., Pantaleoni, F., Fiore, F., Nuschak, B., Peola, S., Foglietta, M., Palumbo, A., Coscia, M., Castella, B., et al. (2005). Effector gammadelta T cells and tumor cells as immune targets of zoledronic acid in multiple myeloma. Leukemia *19*, 664–670.

80. Viey, E., Lucas, C., Romagne, F., Escudier, B., Chouaib, S., and Caignard, A. (2008). Chemokine receptors expression and migration potential of tumor-infiltrating and peripheral-expanded Vgamma9Vdelta2 T cells from renal cell carcinoma patients. J. Immunother. *31*, 313–323.

81. Tu, W., Zheng, J., Liu, Y., Sia, S.F., Liu, M., Qin, G., Ng, I.H.Y., Xiang, Z., Lam, K.T., Peiris, J.S.M., and Lau, Y.L. (2011). The aminobisphosphonate pamidronate controls influenza pathogenesis by expanding a gammadelta T cell population in humanized mice. J. Exp. Med. *208*, 1511–1522.

82. Peters, C., Kouakanou, L., Oberg, H.H., Wesch, D., and Kabelitz, D. (2020). In vitro expansion of Vγ9Vδ2 T cells for immunotherapy. Methods Enzymol. *631*, 223–237.

83. Kim, H.S., Kwon, S.H., Choi, O.K., and Lim, T. (2024). High-dose ascorbic acid synergizes with anti-PD1 therapy in non-small cell lung cancer in vitro and in vivo models. Front. Immunol. *15*, 1512605.

84. Halstead, E.S., Mueller, Y.M., Altman, J.D., and Katsikis, P.D. (2002). In vivo stimulation of CD137 broadens primary antiviral CD8+ T cell responses. Nat. Immunol. *3*, 536–541.

85. Maniar, A., Zhang, X., Lin, W., Gastman, B.R., Pauza, C.D., Strome, S.E., and Chapoval, A.I. (2010). Human gammadelta T lymphocytes induce robust NK cell-mediated antitumor cytotoxicity through CD137 engagement. Blood *116*, 1726–1733.

86. Ye, L., Jia, K., Wang, L., Li, W., Chen, B., Liu, Y., Wang, H., Zhao, S., He, Y., and Zhou, C. (2020). CD137, an attractive candidate for the immunotherapy of lung cancer. Cancer Sci. *111*, 1461–1467.

87. Cairo, C., Armstrong, C.L., Cummings, J.S., Deetz, C.O., Tan, M., Lu, C., Davis, C.E., and Pauza, C.D. (2010). Impact of age, gender, and race on circulating gammadelta T cells. Hum. Immunol. *71*, 968–975.

88. Caccamo, N., Dieli, F., Wesch, D., Jomaa, H., and Eberl, M. (2006). Sex-specific phenotypical and functional differences in peripheral human Vgamma9/Vdelta2 T cells. J. Leukoc. Biol. *79*, 663–666.

89. Burnham, R.E., Zoine, J.T., Story, J.Y., Garimalla, S.N., Gibson, G., Rae, A., Williams, E., Bixby, L., Archer, D., Doering, C.B., and Spencer, H.T. (2020). Characterization of Donor Variability for γδ T Cell ex vivo Expansion and Development of an Allogeneic γδ T Cell Immunotherapy. Front. Med. *7*, 588453.

90. Song, Y., Liu, Y., Teo, H.Y., and Liu, H. (2022). Targeting Cytokine Signals to Enhance γδT Cell-Based Cancer Immunotherapy. Front. Immunol. *13*, 914839.

91. James, K.S., Trumble, I., Clohosey, M.L., Moeser, M., Roan, N.R., Adimora, A.A., Joseph, S.B., Archin, N.M., Hudgens, M., and Soriano-Sarabia, N. (2020). Measuring the contribution of γδ T cells to the persistent HIV reservoir. Aids *34*, 363–371.

92. Kamiya, T., Seow, S.V., Wong, D., Robinson, M., and Campana, D. (2019). Blocking expression of inhibitory receptor NKG2A overcomes tumor resistance to NK cells. J. Clin. Investig. *129*, 2094–2106.

OMTON, Volume *33*

Supplemental information

Targeted expansion of cytotoxic T cells using IL-12 and CD137L supplementation enhances antitumor efficacy

Marta Sanz, Brendan T. Mann, Elyse K. McMahon, Alberto Bosque, Samuel Simmens, and Natalia Soriano-Sarabia

Table S1. Characteristics of the healthy donor volunteers used throughout the study

Donor	Specimen	Sex	Age	Basal Vδ2
1	Healthy	Female	56	0.51
2	Healthy	Male	66	1.17
3	Healthy	Female	49	2.08
4	Healthy	n/a	n/a	2.41
5	Healthy	n/a	n/a	1.57
6	Healthy	Male	45	0.38
7	Healthy	n/a	n/a	1.28
8	Healthy	n/a	n/a	4.62
9	Healthy	Female	44	0.24
10	Healthy	Male	47	0.69
11	Healthy	Female	68	2.46
12	Healthy	Female	59	1.72
13	Healthy	Male	38	2.23
14	Healthy	Female	30	1.35
15	Healthy	Male	60	0.79
16	Healthy	Male	37	0.5
17	Healthy	Female	56	4.51
18	Healthy	Male	31	0.31
19	Healthy	Male	43	2.33
20	Healthy	Female	63	1.42
21	Healthy	Male	33	3.62
22	Healthy	Male	16	12.4
23	Healthy	Female	17	0.8
24	Healthy	Female	17	0.79
25	Healthy	Male	51	0.37
26	Healthy	Male	48	0.047
27	Healthy	Male	41	3.6
28	Healthy	Male	63	1.34
29	Healthy	Female	54	3.75
30	Healthy	Male	38	3.56
31	Healthy	Male	18	0.28
32	Healthy	Female	24	0.16
33	Healthy	Female	18	0.55
34	Healthy	Female	69	0.75
35	Healthy	Male	58	0.59
36	Healthy	Male	41	1.64
37	Healthy	Female	56	0.51
38	Healthy	Male	16	6.61
Mean			43.24	1.95

Participants 20-38 were included in cytotoxicity assays.

Table S2. Median of frequencies upon expansion with N-BPs and IL-2 for 10 days

	Concentration	0	0.08	0.15	0.3	0.6	1.5	2	3	4.5
ALN	Vδ2	1.4	9.4	23.1	31.5	54.9	67.3	64.2	**70.3**	47.0
	Vδ2 CD16	10.6	24.8	29.6	27.8	26.3	26.7	24.7	28.4	25.6
	Vδ2 CD56	6.4	41.3	37.7	40.7	37.0	30.3	29.2	23.2	23.2
	Vδ2 CD8	11.9	12.2	9.8	9.6	8.0	6.9	5.7	5.8	3.7
	Concentration	**0**	**1.8**	**3.6**	**5.3**	**7**	**9**	**18**	**36**	**90**
PAM	Vδ2	1.4	25.6	31.0	32.9	32.3	**33.0**	29.2	18.9	3.1
	Vδ2 CD16	10.6	20.9	16.9	20.0	21.0	22.3	28.3	28.4	33.1
	Vδ2 CD56	6.4	28.4	30.8	25.0	25.3	31.8	25.4	23.4	32.6
	Vδ2 CD8	11.9	7.0	6.3	6.0	6.4	5.6	5.6	7.4	10.1
	Concentration	**0**	**0.5**	**1**	**1.5**	**2.0**	**2.5**	**5.0**	**10**	**25**
ZOL	Vδ2	1.4	44.0	51.1	52.6	53.7	**54.9**	48.7	33.5	11.2
	Vδ2 CD16	10.6	30.2	17.7	27.0	35.2	23.1	28.9	26.2	24.6
	Vδ2 CD56	6.4	23.1	22.5	24.0	23.6	25.5	25.8	22.8	18.9
	Vδ2 CD8	11.9	6.2	5.9	5.2	6.1	5.1	6.8	7.1	7.3

Numbers indicate frequency (%) of CD3+ T cells. N=8

Table S3. Means of the frequencies and mean fluorescence intensity of Vδ2 T cells cultured with different conditions

Experiment	Conditions	**Vδ2**	**Vδ2 CD16**	**Vδ2 CD56**	**Vδ2 NKG2A**	**Vδ2 NKG2D**	**Vδ2 GzmB**	**Vδ2 IFN-γ**	**Viabiliy**	**Vδ2 CD16 GzmB**	**Vδ2 NKG2A GzmB**
1	AIL2	86.86	28.63	30.01	45.30	8011.43	10501.71	2301.00	95.76	10914.71	11140.71
2	AIL2vitC	90.34	31.69	32.87	40.86	7309.57	7595.14	1350.29	97.26	7582.14	7350.71
3	AIL2CD277	91.21	24.46	26.46	35.96	7380.29	8408.86	1423.86	96.49	8169.00	8299.00
4	AIL2CD6	91.80	27.29	30.61	45.09	6214.14	9055.86	2672.86	93.89	9690.00	9578.29
5	AIL2CD137	91.40	28.73	31.36	48.50	6139.43	10576.86	3359.71	90.49	10664.57	10793.57
6	AIL212	78.84	25.61	34.97	49.77	5767.29	14300.43	2016.43	92.06	14927.14	14958.43
7	AIL212vitC	84.23	28.10	42.17	51.27	5574.14	10685.00	1486.71	94.40	10945.71	10848.29
8	AIL212CD277	82.44	21.28	30.70	43.16	5927.14	14321.00	1827.57	90.20	12509.57	14088.57
9	AIL212CD6	84.01	22.55	35.83	51.13	4704.00	13003.86	2806.00	88.63	13271.29	13695.00
10	AIL212CD137	81.17	25.01	36.01	51.51	4674.00	15164.29	3499.86	90.09	18343.71	19708.86
11	AIL221	82.47	25.61	29.89	44.76	6444.00	12427.86	2136.57	91.50	15821.00	15221.43
12	AIL221vitC	87.30	30.17	30.29	47.36	6510.29	10658.00	1465.14	93.44	10889.00	10934.14
13	AIL221CD277	85.57	20.83	25.13	41.69	6823.71	11249.57	1643.71	91.70	10689.86	11472.00
14	AIL221CD6	86.74	25.15	30.26	45.44	4930.57	12337.29	2989.00	88.94	12744.14	12766.29
15	AIL221CD137	86.37	26.02	31.47	46.17	5155.57	12791.29	2948.00	91.16	13473.71	13594.00
	MEAN	86.05	26.08	31.87	45.86	6104.37	11538.47	2261.78	92.40	12042.37	12296.62

Numbers indicate frequency (%) of CD3+ T cells for Vδ2, CD16, CD56, NKG2A and viability, and mean fluorescence intensity (MFI) for NKG2D, GzmB and IFN-γ. N=7

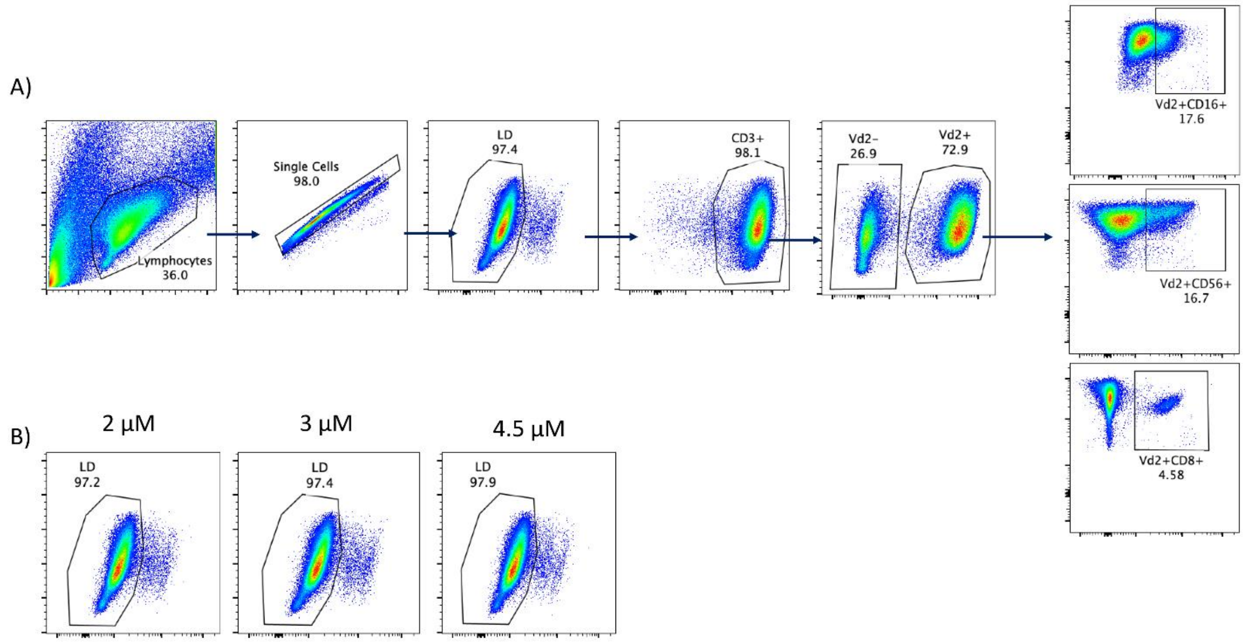

Figure S1: Representative example of the gating strategy

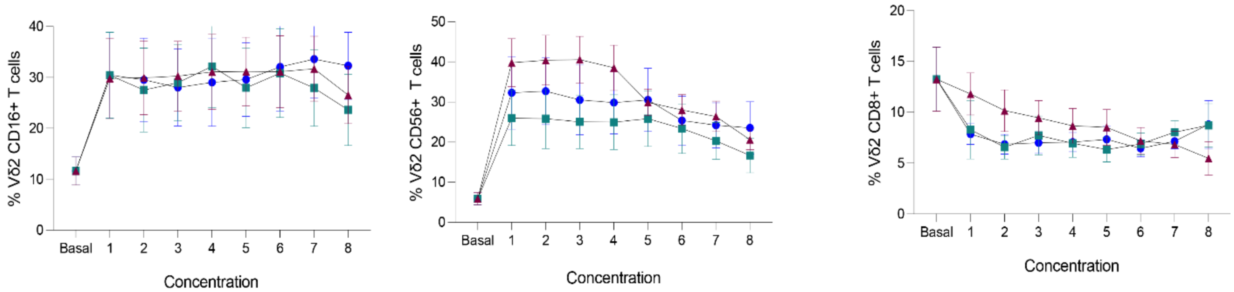

Figure S2: Frequency of Vδ2 T cells expressing CD16 (left), CD56 (middle) or CD8 (right) upon exposure to PAM (blue), ZOL (tile) or ALN (maroon) in cultures of 10 days. N=8.

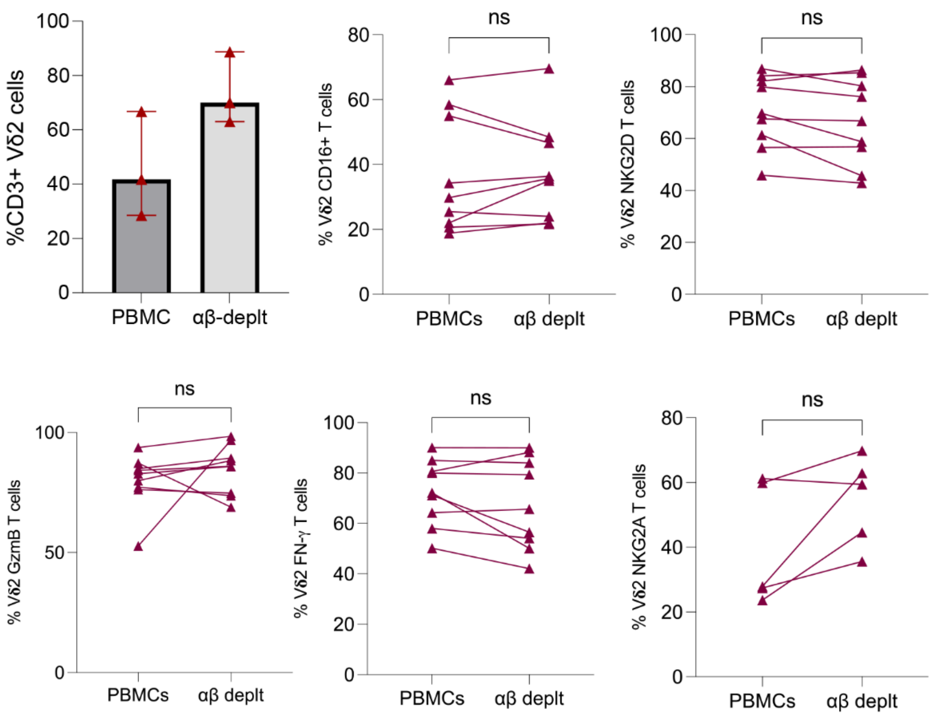

Figure S3: Frequency of total Vδ2 T cells and expression of cytotocix markers on ALN-expanded Vδ2 T cells after 10 days in cultures that started from whole PBMC compared to αβ-TCR+ depleted PBMCs.

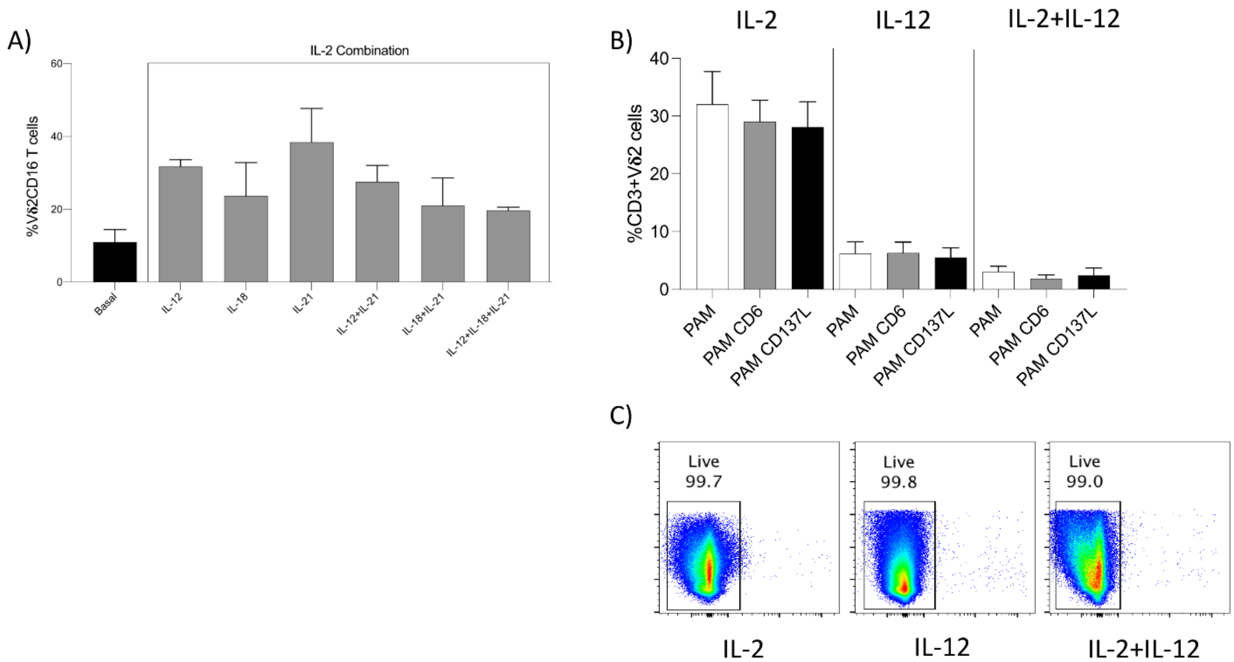

Figure S4: Expansion of VδT cells A) when IL-2 was combined with IL-12, IL-18 and IL-21. B) in the continuous presence of cytokines and UCD6 or CD137L. B) Viability of expanded Vδ2 T cells. N=4

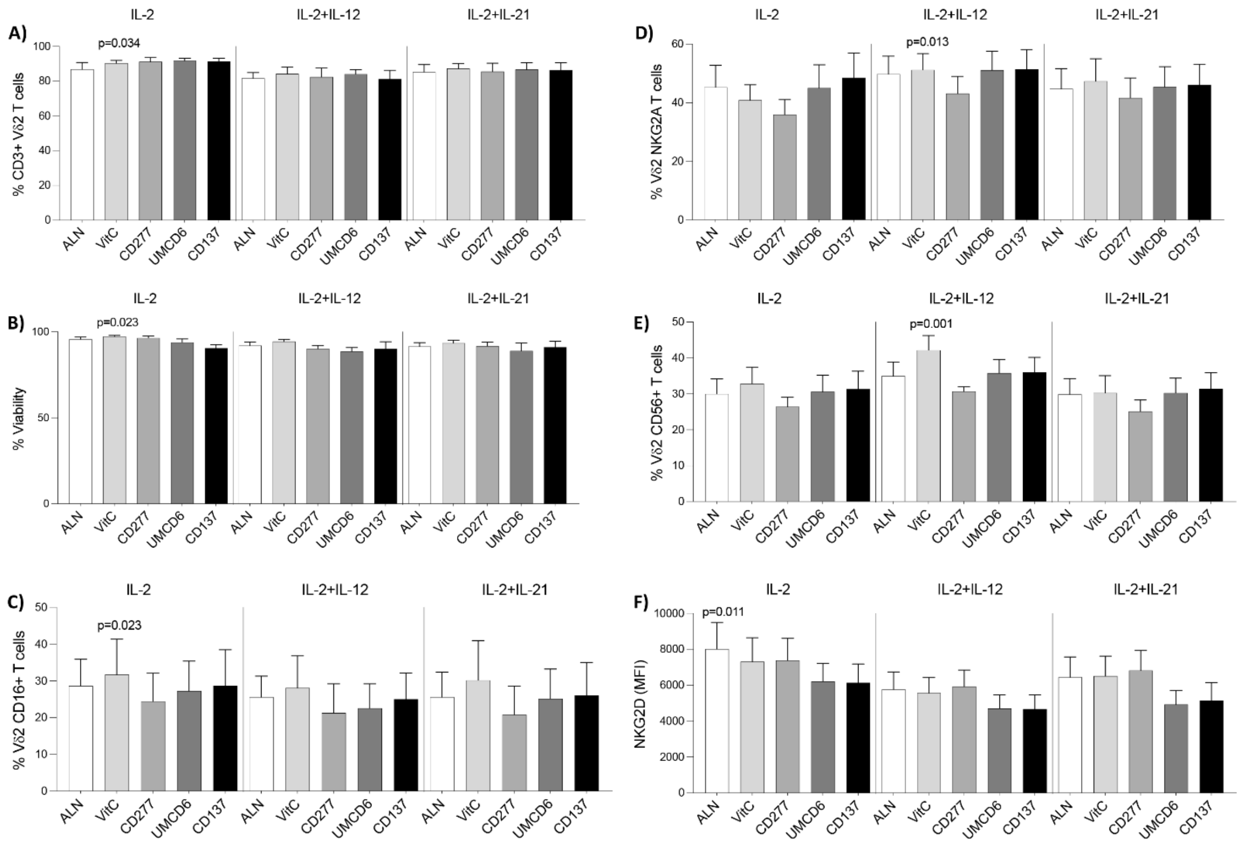

Figure S5: Comparison of Vδ2 T cell expansion after exposure to ALN+IL-2, ALN+IL-2+IL-12 or ALN+IL-2+IL21 combined with either vitamin C, CD277, UCD6 or CD137L. A) Vδ2 frequency, B) viability, and expression of C) CD16, D) NKG2A, E) CD56 and F) NKG2D, in expanded Vδ2 T cells. N=7.

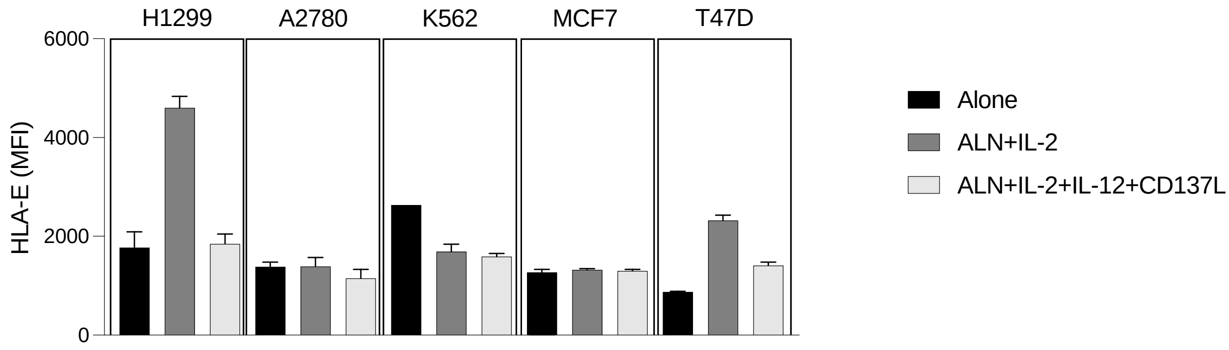

Figure S6: HLA-E expression in the different cancer cell lines